

Context-Triggered Overrides and Geodesic ETA for Online Multi-Robot Task Allocation under Failures, Mixed Stress, and Deadline Pressure

Abstract

A robot team is rarely handed all of its work up front: work appears, expires, and changes hands while the fleet moves. Online multi-robot task allocation must therefore stay sound as tasks arrive, deadlines tighten, and robots fail mid-execution, and no fixed rule serves every regime. We present AHE-MRTA, an event-triggered allocator that chooses online among five interpretable allocation behaviours. A two-level rule makes the choice: deterministic context overrides fire on failure and on deadline pressure, and a dominance dynamic decides when neither fires. Two further mechanisms price the assignment—a geodesic-ETA oracle and a bounded terminal load repair. We evaluate on three layers of realism: allocation-only, a stochastic navigation proxy (500 seeds), and closed-loop Gazebo/ROS 2 Nav2 with 3, 5, and 10 TurtleBot3 robots (480 runs, 20 seeds per cell at the primary scale). AHE-MRTA leads every scenario on all three layers. Under deadline pressure it reaches 0.650 effective on-time completion at 5 robots and 0.764 at 10, against 0.578 and 0.524 for the strongest baseline. It cuts that baseline’s deadline-violation rate by 17% and 49%, and decides in 0.8–3.4 ms at those two scales. At the primary scale, 20 of 63 cross-method comparisons survive Bonferroni correction: 15 in its favour and five against. A pre-registered matched ablation (240 further runs, four arms, 20 common seeds) then separates what carries that result from what does not. Pinning the allocator to a paradigm the context does not call for costs 9–11 percentage points of effective on-time completion (large effect, corrected $p < 0.005$); removing the override cascade changes nothing measurable. An unregistered fifth arm narrows the claim further: pinned to the one paradigm the selector’s fallback returns almost always, the allocator merely matches it (0.638 against 0.650, $p = 0.43$). We therefore claim that the portfolio must hold a paradigm fit for the regime, and that the allocator must not be pinned away from it. We claim neither the particular selector we implemented nor any advantage of it over that fallback.

Index Terms

Multi-robot systems, task allocation, online algorithms, biomimetic optimization, ROS 2, Gazebo.

I. INTRODUCTION

Fielded robot teams are given goals, not schedules. An inspection or delivery mission states what has to be visited and by when; which robot goes where, and in what order, is settled while the mission is already running. That decision is multi-robot task allocation (MRTA). It must respect robot capabilities, task requirements, and operational constraints, and once execution begins it also depends on execution state: a robot may be delayed, unavailable, or committed to a task whose value or feasibility has changed. Recent reviews cover centralized optimization, distributed and market-based methods, metaheuristics, and learning-based policies [1], [2]. Online, then, a low-cost assignment at one instant is not enough: the allocator must preserve useful decisions as the situation moves under it.

No single strategy is best under all of these conditions, because a choice that helps in one regime tends to hurt in another. Three tensions recur. The first is solution quality against reaction cost. Centralized formulations jointly optimize assignment and routing when global state is available, folding timing, workload, distance, energy, and deadline terms into one program [3]. Re-solving that coupled problem at every event is expensive, and it can overturn a plan execution has already begun to act on. The second is global optimality against coordination cost. Distributed and market-based schemes replace the global solve with local decisions and explicit messaging: group-based auctions serve delivery requests that arrive over time with time windows [4], and consensus-based allocation gives up some optimality for local coordination, speed, and lower energy use [5]. The locality that bounds their communication, however, also limits how far they exploit global structure. The third is reactivity against stability. An aggressively re-optimizing policy recovers quickly when a robot fails or a deadline slips, but it preempts running tasks and breaks feasible routes; a commit-once policy is cheap and stable, but it strands the tasks of a failed robot. Other online methods take fixed positions on these axes: game-theoretic clustering couples task grouping with allocation [6], self-adaptive methods react to shifts in problem structure [7], and decentralized energy-aware policies weight power draw [8]. Each commits to its position in advance. Arrival bursts, robot failures, and deadline pressure each reward a different position, which is why we treat the allocation behaviour itself as an online choice rather than a fixed design commitment.

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The method keeps the name AHE-MRTA from its implementation, data directories, and the run identifiers of the released campaign. The name is retained for traceability; the contribution claimed here is the context-override cascade, the geodesic-ETA oracle, and the bounded terminal load repair, not the biomimetic metaphor the acronym originally referred to.

This paper considers online single-task-robot allocation under task arrivals, robot failures, mixed stress, and deadline pressure, where the allocator acts at a task arrival, failure, deadline alarm, or replanning event and may use only the state available then. Reassigning a task can reduce estimated travel cost but can also preempt execution, disrupt a feasible route, or move delay to another robot; keeping an incumbent can be equally harmful after a failure or a sharp loss of time slack. We therefore take completion and deadline violations as primary outcomes, and report delay, workload balance, recovery, redispaches, preemptions, and decision latency as trade-offs.

Our method, *AHE-MRTA*, is an event-triggered allocator, not a new optimizer at every event. Its behaviour is governed by a two-level selector. The upper level is a deterministic override cascade over observable context: when the recent failure rate exceeds a threshold the allocator switches to orphan-first recovery, and when deadline pressure exceeds a threshold it switches to strict earliest-deadline-first dispatch. The lower level is a dominance dynamic over five interpretable behaviours—spatial greedy, priority-first, deadline-oriented temporal, commit-once stability-oriented, and orphan-first recovery—which decides only when neither override fires. Two further mechanisms shape the cost the selected behaviour minimizes: a geodesic-ETA oracle that prices an assignment by traversable map distance, and a bounded terminal load repair that rebalances the closing schedule under a remaining-makespan non-regression guard.

We state at the outset what our own measurements support, including where they contradicted our expectations. A matched closed-loop ablation (Section VII-E) was pre-registered before it was run. It shows that the *portfolio* is load-bearing: pinning the allocator to a paradigm the context does not call for costs 9–11 percentage points of effective on-time completion under deadline pressure, with large corrected effects. It also shows that the selector’s upper layer is not load-bearing. Deleting the override cascade and falling back to $\arg \max_i D_i$ changes nothing measurable, so the two levels are substitutes rather than a mechanism and its ornament. A fifth arm, added after registration and reported as exploratory, marks the other edge of the claim. Pinned to spatial-greedy—the paradigm that fallback returns in 99.9% of the states it decides—the allocator performs like the full selector. We therefore claim online switching over a fixed portfolio, not the specific rule that performs it, and we mark clearly where the evidence stops (Section VII-G). The navigation-proxy ablation separates the variants too, and it agrees with the fifth arm rather than with the registered ones: it puts fixed spatial-greedy ahead of the full selector. That plane charges nothing for congestion or recovery latency, which is exactly what the closed-loop experiment adds.

The paper makes four bounded contributions:

- 1) It specifies a two-level context-override selector for online MRTA that couples fixed context signals to fixed, interpretable allocation behaviours, and quantifies how much each level actually decides.
- 2) It records the decision and execution quantities needed to separate allocation quality from navigation and recovery effects.
- 3) It evaluates the policy across robot-team scales, in a navigation-independent simulator and in a ROS 2/Gazebo stack.
- 4) It reports allocation-only, stochastic navigation-proxy, and closed-loop Gazebo evidence separately, and shows where they agree and where they do not: all three put AHE-MRTA first in every scenario, but the ordering *among* the baselines does not carry across planes—commit-once BiG-MRTA is a consistent runner-up on the proxy and last in two of three Gazebo scenarios—and only the closed loop separates the portfolio (load-bearing) from the selector rule (not).

The three layers differ only in how navigation is modelled—deterministic goal success, stochastic position-dependent goal failure, then the full physics and Nav2 pipeline (Section IV-G)—because a strong score where navigation cannot fail does not by itself demonstrate closed-loop navigation or recovery performance. The closed-loop benchmark compares full methods; the causal effect of the selector is isolated separately, by the matched ablation of Section VII-E.

II. PROBLEM FORMULATION

Event-driven state. Let $\mathcal{R} = \{r_1, \dots, r_n\}$ be the robot set. A requested task τ is described by

$$\xi_\tau = (p_\tau, t_\tau^a, d_\tau, \pi_\tau, c_\tau),$$

where $p_\tau \in \mathbb{R}^2$ is the goal position, t_τ^a is its activation time, $d_\tau \in \mathbb{R}_{\geq 0} \cup \{\infty\}$ is its deadline, $\pi_\tau \in \{1, 2, 3\}$ is its priority, and c_τ is its capability requirement. At event time t_k , $\mathcal{T}_k$ contains every activated task that is not completed or irreversibly cancelled. Robot r has position p_r^k , health state $\phi_r^k \in \{\text{alive}, \text{failed}\}$, navigation status, and an ordered queue q_r^k . The queue contains tasks assigned but not completed, including an in-flight task when one exists.

An allocation event is a task arrival, a robot failure, a deadline alarm, or the replanning timer. Let $\mathcal{R}_k^{\text{av}}$ denote robots that are alive, not navigation-stuck, and have remaining queue capacity. Let $\mathcal{I}_k \subseteq \mathcal{R} \times \mathcal{T}_k$ be the set of healthy in-flight robot–task pairs. These pairs are locked: a new allocation event may change future queue entries but does not cancel an executing Nav2 goal.

Feasible event-level decision. The binary variable $x_{r\tau}^k = 1$ means that task τ belongs to robot r 's planned queue after event t_k . The feasible set is

$$\mathcal{X}_k = \{x \in \{0, 1\}^{n \times |\mathcal{T}_k|} : \sum_{r \in \mathcal{R}} x_{r\tau}^k \leq 1, \quad \forall \tau \in \mathcal{T}_k, \quad (1)$$

$$\sum_{\tau \in \mathcal{T}_k} x_{r\tau}^k \leq Q, \quad \forall r \in \mathcal{R}, \quad (2)$$

$$x_{r\tau}^k = 1 \Rightarrow (c_\tau \subseteq \text{cap}(r)), \quad \forall r, \tau, \quad (3)$$

$$x_{r\tau}^k = x_{r\tau}^{k-1}, \quad \forall (r, \tau) \in \mathcal{I}_k. \quad (4)$$

The first constraint gives a task one owner, the second limits queue length, and the third enforces capability compatibility. The final constraint is the in-flight safety lock used in the execution stack. Failed or navigation-stuck robots cannot receive a new task; their unfinished tasks become eligible for redispach.

Arrival prediction and admissibility. For a candidate pair (r, τ) , let $\hat{a}_{r\tau}^k$ be the predicted arrival time obtained from the current queue endpoint, a queue-length delay surrogate, and travel to p_τ . The predictor uses the cached geodesic ETA of Eq. (20) as its route metric. A pair with no route is infeasible. For a finite deadline, the admission rule is

$$\hat{a}_{r\tau}^k > d_\tau + s_{r\tau}^k \implies (r, \tau) \notin \mathcal{F}_k, \quad (5)$$

where $\mathcal{F}_k$ is the eligible-pair set. The pair-specific allowance $s_{r\tau}^k$ is zero for new tasks in the nominal configuration. A bounded positive allowance is available only to old rescue tasks and can be widened in failure mode. Thus, the allocator rejects a pair that is incompatible, unavailable, unreachable, or later than the allowance for that task state.

Event-level surrogate. Minimizing non-negative cost over $\mathcal{X}_k$ alone would admit the trivial all-zero assignment because the ownership constraints are upper bounds. The implementation therefore uses a cardinality-first rule. After masking infeasible pairs and preserving locked incumbents, define

$$\mathcal{X}_k^* = \arg \max_{x \in \mathcal{X}_k} \sum_{r \in \mathcal{R}} \sum_{\tau \in \mathcal{T}_k} x_{r\tau}^k. \quad (6)$$

Among these maximum-cardinality event plans, the selected paradigm minimizes its shaped cost:

$$x^k \in \arg \min_{x \in \mathcal{X}_k^*} \sum_{r \in \mathcal{R}} \sum_{\tau \in \mathcal{T}_k} C_{r\tau}^k x_{r\tau}^k, \quad C_{r\tau}^k = \infty \text{ if } (r, \tau) \notin \mathcal{F}_k. \quad (7)$$

Repeated LSA passes implement this lexicographic rule; tasks for which no feasible capacity remains stay open for a later event. The active paradigm then sets queue order (Section III-E). This is an online event-level surrogate, not an offline optimum for the complete mission.

Cost and assignment stability. The implementation evaluates seven normalized features: arrival time $A_{r\tau}^k$, priority penalty $P_\tau = (4 - \pi_\tau)/3$, battery risk B_r^k , relative load L_r^k , failure risk F_r^k , deadline urgency U_τ^k , and navigation/recovery risk R_r^k . For a feasible pair, the common cost is

$$\begin{aligned} C_{r\tau}^k &= \rho(\pi_\tau) [w_d A_{r\tau}^k + w_p P_\tau + w_b B_r^k + w_l L_r^k \\ &\quad + w_f F_r^k + w_t U_\tau^k + w_r R_r^k + \Psi_{r\tau}^k], \\ \rho(\pi) &= 1 - 0.1(\pi - 1). \end{aligned} \quad (8)$$

Here $A_{r\tau}^k = \min(1, \hat{a}_{r\tau}^k / A_{\max})$ with $A_{\max} = 220$ s. Because $\hat{a}_{r\tau}^k$ is an absolute time, this feature saturates for every candidate pair once the run passes $A_{\max}$; late in a long run the travel-time channel therefore stops discriminating and the decision is carried by the remaining features and by $\Psi_{r\tau}^k$. In the reported runs (makespans of 150–390 s) this affects the tail of the 10-robot cells. The other base features are dimensionless. The correction $\Psi_{r\tau}^k$ collects the deadline and critical-battery penalties, deadline-capability and bid bonuses, incumbent sticky bonus, reassignment penalty, and capacity penalty. This form avoids adding seconds directly to normalized features. The selector supplies w ; the active paradigm can also shape task order and eligibility. A safe incumbent retains base urgency, whereas an unassigned or at-risk task receives nonlinear urgency growth. The factor ρ was written as a discount for urgent work, and it behaves as one as long as the bracket is positive. In the evaluated configuration the bracket is negative: the deadline-capability bonus inside $\Psi_{r\tau}^k$ outweighs the weighted features. Multiplying a negative bracket by ρ inverts the intent, so ρ charges a priority-3 task *more* than a priority-1 task in the same state. We report the implemented form rather than the intended one, and quantify the effect in Section VII-G.

Run-level outcomes and timing contract. The run-level objective is to maximize completion and minimize deadline violations without hiding unfinished tasks. Let $\mathcal{T}^{\text{done}} \subseteq \mathcal{T}^{\text{req}}$ be the tasks completed by $T_{\max}$. Formally,

$$\max \text{CR} = \frac{|\mathcal{T}^{\text{done}}|}{|\mathcal{T}^{\text{req}}|}, \quad \min \text{DVR} = \frac{|\{\tau : d_\tau < \infty \wedge (t_\tau^c > d_\tau \vee \tau \notin \mathcal{T}^{\text{done}})\}|}{\max(1, |\{\tau : d_\tau < \infty\}|)}. \quad (9)$$

The remaining reported quantities—delay, makespan, recovery time, workload balance, redispaches, preemptions, decision latency, messages, and distance—describe the cost of these outcomes. Delay and DVR are evaluated on all requested tasks; an unfinished task is right-censored at $T_{\max}$ for delay and is a deadline miss for DVR (Section IV-H).

For the deployed AHE cycle, the decision-time contract is $\ell_k = t_k^{\text{publish}} - t_k \leq 50$ ms. It is met per event, not only on average: across the 1,628 allocation decisions logged in the 60 AHE-MRTA runs at the primary scale, the slowest single decision is 4.35 ms and none exceeds the budget. Baseline decision latencies are reported as measured comparison outcomes rather than being silently clipped to this AHE service target. This distinction preserves the real-time requirement while making latency trade-offs visible.

III. AHE-MRTA: A CONTEXT-OVERRIDE ALLOCATOR

Design principle and event-triggered architecture.

Section I argued that no fixed allocation policy is best across changing regimes. AHE-MRTA answers this not with a new global solver but with an event-triggered *selection hyper-heuristic* that chooses, online, among a fixed library of five well-understood low-level behaviours [9]. At each allocation event it observes the fleet state, the open task pool, and the previous queues, updates its context, activates one paradigm, and publishes robot-specific queues. The selector runs no new inter-robot negotiation protocol: the central manager publishes only each robot's own queue, keeping a low-bandwidth interface.

Selection is a two-level cascade, and the levels are not equal partners. The upper level is a pair of deterministic context overrides—one for failure, one for deadline pressure—that fire on threshold crossings and settle three quarters of all events (Section III-B). Only when neither fires does a dominance dynamic over the five behaviours decide (Section III-C); Section VII-G reports how little that tier changes. Two further mechanisms shape the cost the selected behaviour minimizes rather than the choice itself: a cached geodesic-ETA oracle and a bounded terminal load repair (Section III-D). We describe the cascade first because that is where the decisions are made.

The persistent selector state comprises a five-dimensional dominance vector $D_k \in \mathbb{R}_{\geq 0}^5$, the last accepted paradigm, and the remaining dwell counter for that paradigm. The allocator additionally retains in-flight goals, previous task–robot ownership, and an orphan-task pool. This second group is not selector memory: it is needed to enforce the in-flight lock and reassignment safety rules of Section II. Figure 1 summarizes the information flow from an event to queue publication.

A. Context representation

At each allocation event t_k , the observable system state is compressed into the four-dimensional context vector

$$c_k = [c_{1,k}, c_{2,k}, c_{3,k}, c_{4,k}]^T \in [0, 1]^4. \quad (10)$$

Let $n = |\mathcal{R}|$ be the fleet size and $m_k = |\mathcal{T}(t_k)|$ the number of open tasks. Let $\mathcal{R}_k^{\text{alive}}$ denote robots that are neither failed nor navigation-stuck, and $\mathcal{R}_k^f$ its complement. Queue occupancy does not enter $c_{2,k}$: the context is assembled by the ecosystem manager from published robot-status summaries, which carry health and navigation state but not queue length. $\mathcal{R}_k^{\text{alive}}$ is therefore defined by a weaker condition than the allocator-side $\mathcal{R}_k^{\text{av}}$ of Section II, and is a superset of it. The components are

$$c_{1,k} = \min\left(1, \frac{m_k}{\max(1, n)}\right), \quad \text{task density,} \quad (11)$$

$$c_{2,k} = \frac{|\mathcal{R}_k^{\text{alive}}|}{\max(1, n)}, \quad \text{robot availability,} \quad (12)$$

$$c_{3,k} = \frac{|\{\tau \in \mathcal{T}(t_k) : d_\tau - t_k \leq \Delta_d\}|}{\max(1, m_k)}, \quad \text{deadline pressure,} \quad (13)$$

$$c_{4,k} = \frac{|\mathcal{R}_k^f|}{\max(1, n)}, \quad \text{failure rate} \quad (14)$$

Here, $\Delta_d = 60$ s. Tasks with an infinite deadline do not enter the $c_{3,k}$ count. The first two signals describe demand–supply balance, the third the fraction of tasks with shrinking time slack, and the last the capacity loss requiring recovery. They are computed from the task pool and already published robot-status summaries; the selector therefore adds no inter-robot communication burden.

B. Context-override cascade

The acute regimes are handled first and deterministically, without consulting any adaptive state. The raw choice $\tilde{p}_k$ follows the priority order

$$\tilde{p}_k = \begin{cases} \text{H_RECOV}, & c_{4,k} > 0.05, \\ \text{H_TEMP}, & c_{4,k} \leq 0.05 \wedge c_{3,k} > 0.50, \\ \arg \max_i D_{i,k+1}, & \text{otherwise.} \end{cases} \quad (15)$$

TABLE I
CONTEXT PROTOTYPE VECTORS $V_i \in [0, 1]^4$ FOR EACH PARADIGM. COLUMN HEADERS: c_1 TASK-DENSITY (TD), c_2 ROBOT-AVAILABILITY (RA), c_3 DEADLINE-PRESSURE (DP), c_4 FAILURE-RATE (FR). HIGH VALUE = PARADIGM EFFECTIVE WHEN THIS CONTEXT DIMENSION IS HIGH.

Hormone	td	ra	dp	fr
H_SPATIAL	0.7	0.7	0.1	0.1
H_CRIT	0.3	0.5	0.8	0.2
H_TEMP	0.5	0.5	0.9	0.1
H_STAB	0.3	0.3	0.3	0.8
H_RECOV	0.3	0.2	0.2	0.9

Recovery therefore has priority when an event contains both failure and deadline pressure. The third branch is a fallback, described in Section III-C; H_TEMP is used instead whenever D_k is absent or its component range is below 10^{-4} , i.e., when that state is not discriminative.

This ordering is not a presentational choice: it is where the decisions are made. Replaying the selector over the 15,058 logged states of the benchmark campaign, the failure override decides 49.2% of events and the deadline override a further 25.3%, leaving 25.5% to the fallback (Section VII-G reports what the fallback does with them). The two thresholds are the method's operative content.

Because the raw choice can switch rapidly, a change other than H_RECOV is held for four allocation events. More specifically, if $\tilde{p}_k \neq p_{k-1}$ while the dwell counter is positive, the dispatcher returns $p_k = p_{k-1}$ and decrements the counter. Once a change is accepted, the counter is reset to $K_{\text{dwell}} = 4$. We use a distinct symbol here because ρ already denotes the priority factor of Eq. (8). H_RECOV bypasses this hold, so a detected failure does not defer the recovery pass to the next event. This mechanism retains override reactivity while limiting ping-pong switching in normal regimes.

C. Cost weights and the dominance fallback

Each paradigm $i \in \{1, \dots, 5\}$ has a designer-defined context prototype $V_i \in [0, 1]^4$. The prototypes are not learned class centres; they are fixed, interpretable priors on which signal combination supports the associated behaviour. Compatibility with the current context is

$$v_{i,k} = \frac{V_i^\top c_k}{\|V_i\|_2 \|c_k\|_2} \quad \text{and} \quad \mathbf{v}_k = [v_{1,k}, \dots, v_{5,k}]^\top, \quad (16)$$

with zero compatibility if either vector has zero norm. Table I gives the prototypes.

a) *Dominance fallback.*: When neither override fires, the third branch of Eq. (15) reads a five-dimensional dominance vector D_k , initialized uniformly and kept on the probability simplex at every event ($D_0 = \frac{1}{5}\mathbf{1}$, $D_k \succeq 0$, $\|D_k\|_1 = 1$). Its update

$$D_{k+1} = \text{Proj}_{\Delta^4} \left(\text{clip}_{[0,1]} [\alpha D_k + \eta A D_k - \lambda S D_k + \beta \mathbf{p}_k + \gamma \mathbf{v}_k + \delta \mathbf{b}_k] \right) \quad (17)$$

combines interaction-inspired winner-reinforcement (A) and competitor-suppression (S) terms with context compatibility $\mathbf{v}_k$ (Eq. (16)), short-horizon performance $\mathbf{p}_k = g_k \mathbf{v}_k$ scaled by the completed-minus-failed ratio $g_k = (N_k^c - N_k^f) / \max(1, N_k^c + N_k^f)$, and a failure support $\mathbf{b}_k = c_{4,k}[-0.3, 0, 0, 0.4, 0.6]^\top$ that lifts H_RECOV and H_STAB while damping the purely spatial behaviour [10]. Here Proj_{Δ^4} is the implementation's non-negative ℓ_1 normalization with a uniform fallback, not an iterative Euclidean projection. The interaction matrices are nearly empty: A has two nonzero entries (H_CRIT supports H_TEMP and H_STAB supports H_RECOV, both 0.20) and S has one (H_TEMP suppresses H_SPATIAL at 0.30); every other entry is zero. Functionally this is a fixed, low-dimensional smoothing of paradigm preference that supplies learning-free hysteresis. It is not an online learned model, carries no optimality guarantee, and—as Section VII-G quantifies—changes the selected paradigm in three of the 15,058 replayed states. We report it because the system runs it, not because the evidence supports it as a contributor. The same vector does carry weight generation below, which is load-bearing.

b) *Weight generation.*: The selector also produces the seven weights of the common cost function. The channels correspond, in order, to travel time, priority, battery risk, queue load, failure risk, deadline urgency, and reassignment/recovery cost. The fixed paradigm-weight map is

$$M = \begin{bmatrix} 0.9 & 0.1 & 0.1 & 0.3 & 0.3 \\ 0.1 & 0.9 & 0.5 & 0.5 & 0.1 \\ 0.1 & 0.1 & 0.1 & 0.3 & 0.3 \\ 0.1 & 0.1 & 0.1 & 0.1 & 0.3 \\ 0.1 & 0.1 & 0.1 & 0.9 & 0.9 \\ 0.1 & 0.5 & 0.9 & 0.1 & 0.1 \\ 0.1 & 0.1 & 0.1 & 0.3 & 0.9 \end{bmatrix}. \quad (18)$$

Accordingly,

$$w_k^{\text{eco}} = \text{softmax}\left(\frac{MD_{k+1}}{T_w}\right), \quad w_k = 0.3w_0 + 0.7w_k^{\text{eco}}, \quad (19)$$

where $T_w = 0.3$. When recovery mode is active, w_k is additionally blended toward w_{rec} with $\theta_k = \min(0.80, 0.50 + 0.60c_{4,k})$. If deadline pressure exceeds 0.5 and recovery mode is inactive, the urgency channel is multiplied by 1.5 and clipped at 1. These weights are the context-dependent part of the common cost in Section II. The evaluated execution stack has no battery model, so the battery *feature* is identically zero in every reported run and the channel contributes nothing to any cost, even though its weight w_b is not zero (Section III-G). Channel and weight are kept in the cost vector so that the same weight map transfers unchanged to battery-aware deployments.

c) *Feasibility and stability.*: Every paradigm applies the hard admission gate of Eq. (5) and the single-owner and in-flight constraints. For unassigned or at-risk tasks that have consumed more than 0.7 of their deadline budget, the urgency penalty is increased nonlinearly. A task expected to finish on time with its incumbent owner is exempt. Changing a previous owner requires at least a 10% improvement in estimated arrival; three classes are exempt from this margin rule—orphans whose previous owner is no longer healthy, rescue tasks, and tasks already inside the urgent horizon—so the gate never delays work that is failing or about to expire. These safeguards prevent event-triggered replanning from becoming gratuitous task migration.

D. Geodesic ETA and bounded load repair

AHE-MRTA uses a cached geodesic cost oracle and a terminal, bounded load-repair step. Neither component changes the selector or the paradigm library. The cost oracle uses cached geodesic distance d_G on the 0.55 m-inflated occupancy mask of the shared Gazebo map, using an eight-connected free-cell graph. A pair is infeasible if no route exists. For robot r with ordered queue $Q_r = (q_1, \dots, q_j)$, let $e_r(Q_r) = p_r$ when $j = 0$ and $e_r(Q_r) = p_{q_j}$ otherwise. The implemented ETA surrogate is

$$\hat{a}_{r\tau}^k = t_k + \frac{d_G(e_r(Q_r), p_\tau)}{v_r} + j(h + s_\tau), \quad (20)$$

where s_τ is the candidate service time and $h = 22$ s is the fixed per-queued-task navigation allowance. The speed $v_r = 0.22$ m/s is the *effective* traversal rate, not the platform maximum $v_{\text{max}} = 0.26$ m/s of Section IV-A: it absorbs planning and acceleration overhead, and it is the value the implementation divides by. This deliberately inexpensive surrogate does not claim to reconstruct every intermediate route edge. The same endpoint distance oracle is used by the admission gate, paradigm costs, and repair burden calculations.

After the primary plan x^0 is produced, the bounded load-repair step considers local candidates that change the robot only for newly assigned tasks. It therefore never cancels an executing Nav2 goal. Let $M(x)$ be the predicted number of deadline misses, $D_G(x)$ total geodesic distance, $J(x)$ the projected active-robot Jain index, $B_r(x)$ accumulated and committed productive load, and $F(x)$ the maximum predicted remaining completion time. The admissible candidate set is

$$\begin{aligned} \mathcal{N}(x^0) = \{x : M(x) \leq M(x^0), \quad D_G(x) \leq (1 + \epsilon)D_G(x^0), \\ J(x) \geq J(x^0), \quad \max_r B_r(x) \leq \max_r B_r(x^0), \\ F(x) \leq F(x^0)\}, \quad \epsilon = 0.10. \end{aligned} \quad (21)$$

Within this set, candidates are ranked by first increasing $J(x)$ and then, in order, decreasing maximum load, load range, and total distance. Repair is enabled only in the terminal phase, when the remaining active-task count is at most three times the healthy-robot count. For an executing first task, $F(x)$ can use Nav2's published remaining-path length instead of a static grid query. These guards prevent the repair from accepting a visually more balanced reassignment that increases deadline misses, route cost, or remaining makespan.

The geodesic ETA and bounded repair change execution cost but leave the selector unchanged; the closed-loop benchmark therefore does not independently test the causal effect of the selector. This distinction is maintained in the interpretation of the results.

E. Paradigm library

The five behaviours share the same feasible set and cost infrastructure; they differ in task-processing order, cost shaping, and treatment of incumbent ownership. Given a feasibility-masked cost matrix $C_k^{(p)}$, the common assignment kernel is

$$x_k^{(p)} \in \arg \min_{x \in \mathcal{X}_k^*} \sum_{r \in \mathcal{R}} \sum_{\tau \in \mathcal{T}(t_k)} C_{r\tau,k}^{(p)} x_{r\tau}. \quad (22)$$

For an infeasible robot–task pair, $C_{r\tau,k}^{(p)} = \infty$. In the implementation, linear sum assignment (LSA) is repeated across task tiers or processing phases. Equation (22) is therefore the surrogate for one assignment pass, not a claim of global optimality for the entire run.

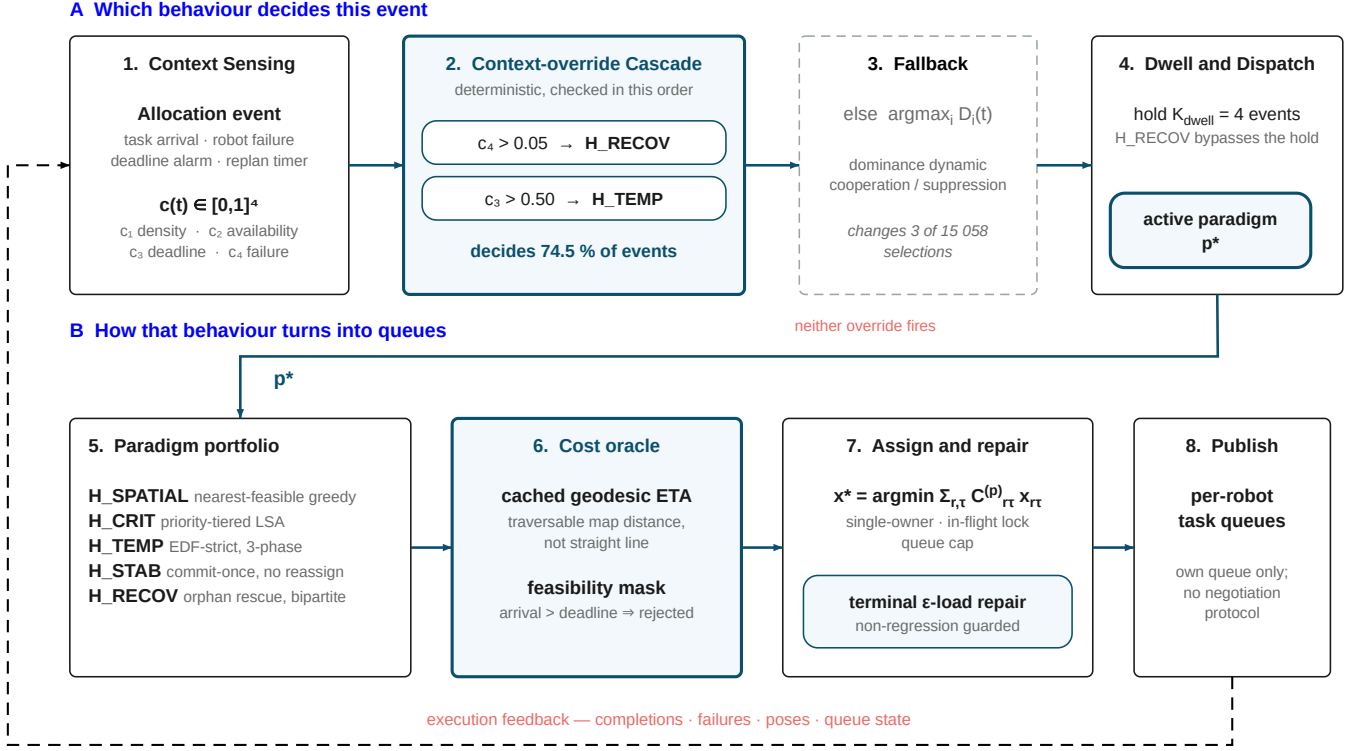


Fig. 1. One allocation event, end to end. **Row A** decides *which* behaviour handles the event. Context sensing (1) compresses the state into $c(t)$; the deterministic override cascade (2) is checked in the order shown and settles most events on its own; only when neither threshold fires does the dominance fallback (3) choose; dwell hysteresis (4) then publishes the active paradigm p^* . The two tiers are not equal partners: replayed over the 15,058 logged ecosystem states of the 120 AHE-MRTA benchmark runs, the cascade decides 74.5% of events, and replacing the fallback with a constant changes 3 of those selections (Section VII-G)—which is why the fallback is drawn muted. **Row B** turns p^* into queues: the portfolio (5) fixes task order, the cached geodesic-ETA oracle (6) prices and masks the pairs, the assignment (7) is taken under single-owner and in-flight locks and closed by the terminal bounded load repair, and (8) publishes per-robot queues. Execution feedback re-enters at (1). Fixed coefficients are listed in Section III-G.

TABLE II
FIVE HORMONES, FIVE ALLOCATION PARADIGMS.

Hormone	Paradigm	Inner mechanism
$H_SPATIAL$	spatial_greedy	nearest-feasible + reject
H_CRIT	priority_first	priority-tiered LSA
H_TEMP	edf_strict	EDF bipartite (multi-phase)
H_STAB	commit_once	hard sticky, no reassign
H_RECOV	orphan_first	orphan rescue + bipartite

- **$H_SPATIAL$ / *spatial_greedy***: Tasks are processed in deadline order and assigned to the eligible robot with the smallest expected arrival time. This follows the nearest-feasible rule used in metric MRTA [11], [12].
- **H_CRIT / *priority_first***: Tasks are partitioned into priority tiers $\pi_\tau = 3, 2, 1$. LSA is applied first to the most critical tier and then to lower tiers, so critical tasks claim capacity first. The design follows the tiered allocation idea in priority-aware auction and consensus strategies [13], [14].
- **H_TEMP / *edf_strict***: The default temporal behaviour processes tasks in increasing d_τ order and applies the hard deadline gate. The implementation uses a fast pass for near-deadline tasks, a recovery pass for any current orphans, and an LSA pass for the remaining tasks. Queue order may be improved by cheapest-insertion candidates that preserve deadline feasibility. The behaviour applies EDF to deadline-constrained MRTA [15], [16].
- **H_STAB / *commit_once***: Tasks with a previous owner that remains eligible are kept with that owner. Only new tasks receive a single-pass allocation. This lowers reassignment churn but limits flexibility after a failure, making the stability trade-off explicit.
- **H_RECOV / *orphan_first***: Tasks left by failed, stuck, or navigation-aborted robots are first redistributed among healthy

Algorithm 1 AHE-MRTA Event Cycle

Require: $\mathcal{R}, \mathcal{T}(t_k)$, previous queues, D_k, p_{k-1} , dwell counter h_k ; fixed A, S, V, M

```

1:  $c_k \leftarrow \text{CONTEXTVECTOR}(\mathcal{R}, \mathcal{T}(t_k))$ 
2: Update the feasibility mask, orphan pool, and in-flight locks
3:  $\mathbf{v}_k \leftarrow (\cos(V_i, c_k))_{i=1}^5$ ;  $\mathbf{p}_k, \mathbf{b}_k \leftarrow \text{PERFORMANCEANDFAILURESSUPPORT}()$ 
4:  $D_{k+1} \leftarrow \text{UPDATEDOMINANCE}(D_k, \mathbf{v}_k, \mathbf{p}_k, \mathbf{b}_k, A, S)$  {Eq. (17)}
5:  $w_k \leftarrow \text{GENERATEWEIGHTS}(D_{k+1}, c_k, M)$  {Eq. (19)}
6:  $\tilde{p}_k \leftarrow \text{RAWCHOICE}(c_k, D_{k+1})$  {Eq. (15)}
7:  $(p_k, h_{k+1}) \leftarrow \text{APPLYDWELL}(\tilde{p}_k, p_{k-1}, h_k)$ 
8:  $x_k \leftarrow \text{PARADIGM}_{p_k}(\mathcal{R}, \mathcal{T}(t_k), w_k)$  {Table II}
9: if in the terminal repair phase then
10:    $x_k \leftarrow \text{EPSILONLOADREPAIR}(x_k, d_G, \epsilon)$ 
11: end if
12:  $x_k \leftarrow \text{SINGLEOWNERANDINFLIGHTLOCK}(x_k)$ 
13: Publish robot-specific queues and store  $(D_{k+1}, p_k, h_{k+1})$ 

```

robots through LSA; remaining tasks then receive temporal allocation. Recovery therefore obtains a dedicated decision pass before new work [12], [17].

F. Algorithm

Algorithm 1 gives the execution order at one allocation event. A task arrival, robot failure, deadline alarm, or replanning timer triggers the cycle. Only queues are published at the end: D_k, A, S, V , and the full cost matrix remain at the manager node.

G. Configuration

The same selector settings are used at all scenarios and fleet scales: $\alpha = 0.65, \beta = 0.40, \gamma = 0.20, \eta = \lambda = 0.12, \delta = 0.20, T_w = 0.3$, and $K_{\text{dwell}} = 4$. The base cost profile is $w_0 = (0.34, 0.10, 0.04, 0.16, 0.10, 0.22, 0.04)$ and the recovery profile is $w_{\text{rec}} = (0.55, 0.04, 0.03, 0.22, 0.09, 0.05, 0.02)$, in the channel order given under *Weight generation*. These quantities are neither relearned nor retuned by event or scenario.

Context, dominance, override, and dwell updates have fixed dimension and are therefore $O(1)$. The dominant cost is the LSA call within a paradigm, whose worst-case complexity for one square assignment pass is $O(\max(n, m_k)^3)$. The geodesic cache reduces repeated goal queries, and load repair is restricted to a fixed number of local move passes. Computational load thus stems from the selected paradigm and optional route/repair layers rather than from the selector itself.

IV. EXPERIMENTAL DESIGN

A. Hardware Platform

All experiments run on an Intel Core i7-11800H workstation with eight physical cores, 16 threads, a 2.30 GHz base clock, 16 GiB DDR4 RAM, Ubuntu 24.04 LTS, and Linux 6.17. The machine has an NVIDIA RTX 3050 Mobile GPU, but no proprietary driver is installed. Gazebo and RViz therefore use Mesa llvmpipe software rasterization (`LIBGL_ALWAYS_SOFTWARE=1, GALLIUM_DRIVER=llvmpipe`). No experiment uses GPU acceleration. The reported comparison is CPU-bound.

B. Software Stack

We use **Gazebo Harmonic** (gz-sim 8.x) [18], **ROS 2 Jazzy** [19], and **Nav2** [20]. Each robot is a **TurtleBot3 Waffle Pi** [21] with a 360-ray two-dimensional lidar (5 Hz, 8 m range), a 200 Hz IMU, and a differential-drive controller ($v_{\text{max}}=0.26$ m/s, $\omega_{\text{max}}=1.82$ rad/s).

Each robot has an independent Nav2 lifecycle in its own ROS 2 namespace. The stack uses global and local costmaps with a 0.40 m inflation radius, a Regulated Pure-Pursuit controller, and a behavior-tree navigator. We provide ground-truth localization through a static `map→odom` transform anchored at each robot's spawn pose. This is an experimental control: it isolates allocation effects from localization error (Section VII-G). The `ros_gz_bridge` relays `/scan`, `/odom`, `/cmd_vel`, `/joint_states`, and TF for each robot. A TF relay aggregates the namespaced trees for visualization. All methods use identical Nav2 parameters. Figure 2 shows how the ecosystem manager and the per-robot Nav2 lifecycles are wired together in this stack.

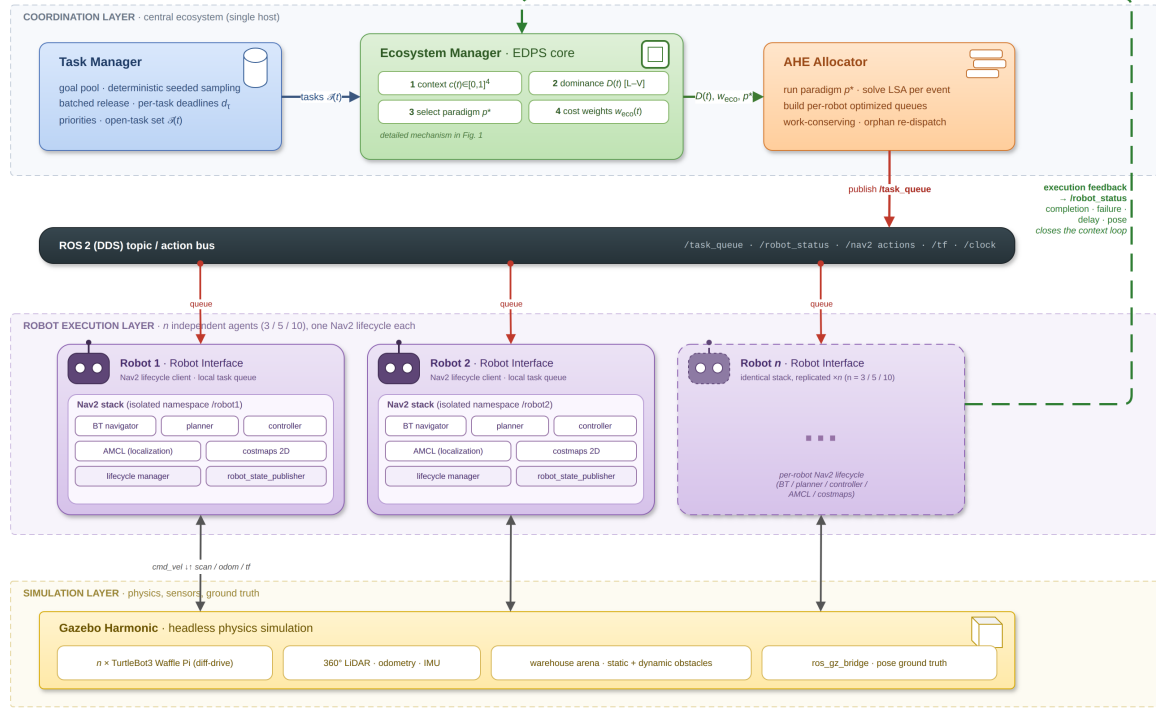


Fig. 2. System architecture. The ecosystem manager updates the hormone state, selects the active paradigm, and publishes one task queue per robot through ROS 2 topics. Each robot runs its own Nav2 lifecycle and returns completion, failure, and delay feedback to the manager.

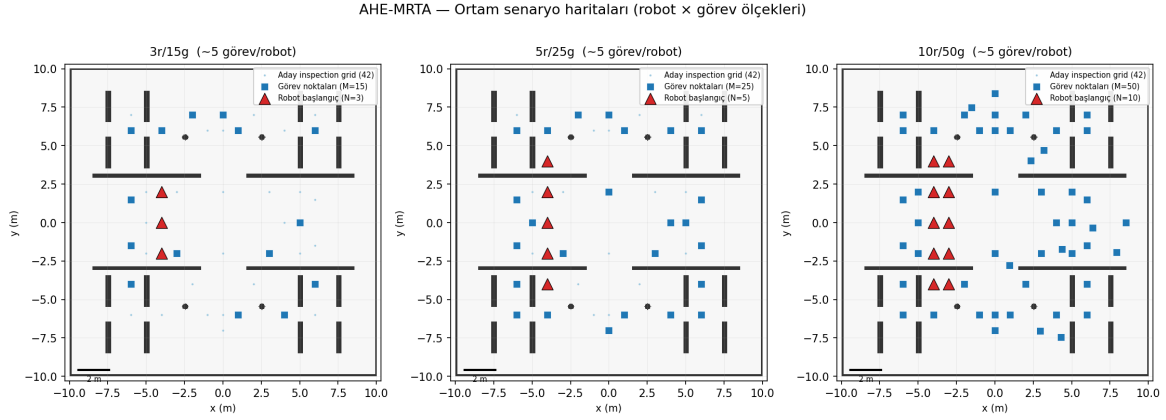


Fig. 3. Arena configurations by scale. **Left:** 3r/15t, with robots in one central column. **Center:** 5r/25t, with two spawn columns. **Right:** 10r/50t, with all five columns active. Blue stars are task goals for seed 1; colored triangles are robot spawns. All waypoints are obstacle-free.

a) Evaluated configuration and artifacts.: AHE-MRTA* is evaluated with its geodesic-ETA oracle and terminal load repair active. Both are switched on explicitly at launch: the allocator’s built-in defaults reproduce the earlier Euclidean, repair-free reference kept for ablation, so the configuration—not only the method name—determines what runs. The resolved switch set is recorded in each run’s metadata and in the campaign directory, so a run cannot be confused with a differently configured one under the same name. The allocator, the campaign drivers, and the scripts that generate every table and figure from the raw logs accompany this submission.

C. Inspection Arena

The arena is a 20×20 m walled indoor mock-up. Two horizontal walls with a 1.5 m central passage create upper and lower lanes. Two rows of square pillars (0.5×0.5 m) line the side walls, and four cylindrical obstacles ($r=0.35$ m) lie in the lanes. We define 42 fixed inspection waypoints. The waypoints are obstacle-aware and at least 1 m apart. Each run samples its task set deterministically from this grid using its seed. Figure IV-C shows the arena, spawn positions, and sampled task sets.

D. Evaluation Scales

We evaluate three Gazebo scales (Table III). Each scale–density cell contains 60 runs (4 methods $\times$ 3 scenarios $\times$ 5 seeds), except the primary 5-robot cell, which carries 20 seeds per method–scenario combination for 240 runs.

TABLE III
GAZEBO EVALUATION SCALES

Scale	Robots	Tasks	Density	Total exps
3r-low	3	9	3.0 t/r	60
3r-mid	3	15	5.0 t/r	60
3r-high	3	24	8.0 t/r	60
5r-mid	5	25	5.0 t/r	240
10r-mid	10	50	5.0 t/r	60
Total Gazebo experiments				480

At $N > 3$, some per-robot `odom`→`base_link` transforms arrived after costmap activation. We resolve this lifecycle race by delaying activation for robot i by $6i$ s at 3r and 5r and by $20i$ s at 10r. We also broadcast an identity transform at node startup. All scales use the same arena, obstacle map, and Nav2 parameters.

E. Stress Scenarios

The three scenarios stress different allocation requirements:

Every run has the same horizon $T_{\max} = 900$ s; a task still open at $T_{\max}$ is unfinished (Section IV-H). In all three scenarios each task draws an independent priority $\pi_\tau \sim \mathcal{U}\{1, 2, 3\}$, a service time $\sim \mathcal{U}[2, 8]$ s, and a deadline $d_\tau = t_0 + \Delta_\tau$ measured from the run start t_0 ; the scenarios differ in the deadline budget Δ_τ , the task-arrival schedule, and whether a robot fails.

- **robot_failure**: all tasks are released at t_0 with a nominal budget $\Delta_\tau \sim \mathcal{U}[90, 300]$ s. One designated robot is disabled at $t = 45 \pm 5$ s (uniform jitter, drawn per seed), early enough that it is carrying work. Its assigned tasks become orphaned and must be redistributed. This scenario tests reactive recovery and the orphan-rescue behaviour.
- **mixed_stress**: the same failure event is combined with staggered demand and a tighter budget: tasks arrive in three waves at $t = 0, 30$, and 60 s ($\lfloor M/3 \rfloor$, $\lfloor M/3 \rfloor$, and the remainder), and the budget is halved ($\Delta_\tau \sim \mathcal{U}[45, 150]$ s). Because deadlines run from t_0 , later waves arrive with less slack, so the allocator must absorb arrivals, priority spread, and a failure at the same time.
- **deadline_pressure**: no robot fails and all tasks are released at t_0 , but the deadline budget is cut to 40% ($\Delta_\tau \sim \mathcal{U}[36, 120]$ s). The lower end of this range is below the Nav2 travel time to the farthest waypoint at $v_{\max}$, so part of the set is unservable on time by construction and the scenario separates urgency-driven dispatch and EDF ordering.

Failure injection targets a fixed robot index rather than a random one, so that the recovery burden is identical across methods for a given seed; task positions, priorities, service times, deadline draws, and the injection time are all seed-controlled and shared by all four methods.

F. Baseline Methods

We compare one operational adaptation from each online-MRTA family represented in the AHE-MRTA library. All adaptations run in the same framework. They share the event loop, ROS 2 topic interface, cost inputs (arrival time, urgency, priority, and load), and Nav2 client. Only the allocation policy differs. The per-robot queue cap Q of Eq. (2) is part of that policy rather than a shared constant: BiG-MRTA and Consensus-DBTA hold $Q=5$, RoSTAM-EA is unbounded within the event, and AHE-MRTA deliberately runs shallow at $Q=2$ (raised by one in the terminal repair phase) so that queued work stays reassignable after a failure. We use source-paper parameters where they are specified and otherwise report fixed benchmark choices. The names below thus identify our implementations, not independent reproductions of the complete source systems.

a) *BiG-MRTA* [22] (*bipartite matching*).: Our BiG-MRTA adaptation builds a bipartite robot–task graph with fixed multi-criteria edge utility and solves maximum-weight matching at each event. Once assigned, a task is not reallocated. It is the deterministic, lowest-latency, single-paradigm comparison. Its policy avoids replanning but does not recover a task when its assigned robot fails or time slack collapses.

b) *RoSTAM-EA* [7] (*evolutionary*).: Our RoSTAM-inspired EA evolves candidate assignment vectors with tournament selection, crossover, and mutation at each allocation event. It dispatches the elite individual. It represents self-adaptive stochastic search. It searches a richer assignment space than one-shot matching, but its decision latency is two to three orders of magnitude higher. Its reoptimization does not encode the cause of a context change.

c) *Consensus-DBTA [5] (DBTA2-style consensus bidding)*.: Each robot shares its two highest-valued task bids; simulated network-wide maximum consensus identifies a winner for each task, and priority rounds prevent one robot from winning conflicting tasks. Its bidding rounds keep assignments largely stable, though it still redispaches after a failure. It represents distributed auction/consensus and is the strongest completion-rate baseline in this study. Re-bidding rounds can delay response to a failure or rapid deadline escalation.

The external baselines are not substitutes for a selector ablation: they differ from AHE in allocation logic as well as paradigm choice. Two separate ablations address the selector question directly—one on the proxy plane (Section VII-D) and a pre-registered matched one in the closed loop (Section VII-E).

d) *Comparison scope*.: The baseline implementations share the event loop, interfaces, and cost inputs, but they are operational re-implementations rather than independent replications of every source system. The four-method benchmark therefore compares these implementations in one common stack. It is not a universal ranking of the cited algorithms. Attributing a closed-loop difference to the selector itself requires a controlled Gazebo ablation against fixed paradigms; that experiment was pre-registered and run, and Section VII-E reports what it does and does not support.

G. Evaluation settings

We use three evaluation settings that differ only in how navigation is modelled, so that allocation quality is never conflated with navigation or recovery behaviour. (i) **Navigation-independent** (allocation-only): every dispatched goal is reached deterministically along the shared geodesic map—no local-planner stall, costmap rebuild, or goal failure—and the robot pose advances to each completed goal, while injected robot failures still occur. This setting isolates assignment and queue-ordering quality from navigation noise. (ii) **Stochastic navigation proxy**: navigation cost is a fixed edge, but a goal can fail position-dependently and must be re-attempted, so the metric also rewards resilience to navigation failure. (iii) **Closed-loop Gazebo/ROS 2**: the full physics and Nav2 pipeline above runs end to end. Results from the three settings are tabulated separately and never pooled.

H. Metrics

We report completion rate (CR), deadline violation rate (DVR), makespan, all-task average delay, failure-recovery time, workload balance (Jain), task redispach rate, execution preemptions, decision latency, and message count. We also report total travel distance where applicable. CR and DVR are the primary outcomes. The other metrics describe efficiency and trade-offs.

a) *All-task delay and DVR*.: A completed-only average rewards a policy that abandons difficult work. Distant, late, or contended tasks then disappear from the denominator. We avoid this survivorship bias by scoring delay and DVR over *all* requested tasks. Let $\mathcal{T}$ be the requested set, t_τ^a the activation time, t_τ^c the completion time, and $T_{\max}$ the run horizon. Delay for an unfinished task is right-censored at the horizon:

$$\delta_\tau = \begin{cases} t_\tau^c - t_\tau^a, & \tau \text{ completed,} \\ T_{\max} - t_\tau^a, & \tau \text{ unfinished,} \end{cases} \quad \bar{\delta} = \frac{1}{|\mathcal{T}|} \sum_{\tau \in \mathcal{T}} \delta_\tau, \quad (23)$$

Any task with a deadline is a violation if it completes late *or* never completes:

$$\text{DVR} = \frac{|\{\tau : d_\tau < \infty \wedge (t_\tau^c > d_\tau \vee \tau \text{ unfinished})\}|}{\max(1, |\{\tau : d_\tau < \infty\}|)}. \quad (24)$$

We apply this convention to every method and scale. We also record completed-only delay and DVR. The two definitions coincide when CR=1 and differ only when a method leaves tasks unserved.

I. Statistical Analysis

We use pairwise, two-sided Mann–Whitney U tests with Bonferroni correction within each scenario family (three baselines $\times$ seven metrics, or 21 tests per scenario). At the primary 5-robot scale, each cell has $n=20$ runs (one density and 25 tasks). The 3-robot scale pools three densities ($n=15$), and the 10-robot scale has $n=5$. We also report Cliff’s δ , an effect-size estimate suited to small cells. The 10-robot Gazebo cells, at $n=5$, are descriptive. We make no cross-method superiority claim from an uncorrected or non-significant comparison. The 100-seed-per-cell allocation-only campaign and the 500-seed primary navigation-proxy cells are separate controlled tests. Gazebo results describe behavior in the evaluated execution stack, not general deployment performance.

V. ALLOCATION-ONLY AND NAVIGATION-PROXY RESULTS

In the navigation-independent setting (Sec. IV-G), the allocator and execution oracle both use geodesic distance on the same inflated occupancy map, so the metrics measure allocation, queue order, and recovery after failure rather than local-planner noise.

Table IV reports AHE-MRTA across the three scales (100 seeds per cell). Completion is exact—CR=1.000 in all nine scale-scenario cells—but fitness is not, because a task that finishes after its deadline earns no credit even when nothing is

TABLE IV
AHE-MRTA NAVIGATION-INDEPENDENT ALLOCATION-ONLY RESULTS (100 SEEDS PER CELL). NAVIGATION SUCCEEDS DETERMINISTICALLY;
DISTANCE IS THE GEODESIC PATH LENGTH ON THE SHARED INFLATED OCCUPANCY MAP.

Scale	Scenario	Fit.	CR	DVR	Jain _{active}	Jain _{dist}	Delay (s)	Distance	Decision (ms)
3r/15t	Robot failure	0.968	1.000	0.036	0.994	0.841	84.3	148.6	0.066
3r/15t	Mixed stress	0.647	1.000	0.353	0.992	0.828	55.5	142.7	0.068
3r/15t	Deadline pressure	0.774	1.000	0.232	0.986	0.965	62.3	119.3	0.070
5r/25t	Robot failure	0.994	1.000	0.005	0.988	0.900	71.5	222.4	0.082
5r/25t	Mixed stress	0.727	1.000	0.271	0.983	0.886	40.7	201.6	0.088
5r/25t	Deadline pressure	0.835	1.000	0.164	0.985	0.960	57.0	172.8	0.090
10r/50t	Robot failure	1.000	1.000	0.000	0.988	0.923	64.8	408.4	0.140
10r/50t	Mixed stress	0.782	1.000	0.215	0.986	0.905	31.7	333.1	0.215
10r/50t	Deadline pressure	0.888	1.000	0.116	0.987	0.945	53.5	303.6	0.169

TABLE V
ALLOCATION FITNESS (PRIORITY-WEIGHTED ON-TIME COMPLETION), 500 SEEDS, 5 ROBOTS / 25 TASKS. **LEFT:** PERFECT-NAVIGATION PLANE, WHICH ISOLATES ALLOCATION QUALITY. **RIGHT:** THE STOCHASTIC NAVIGATION PROXY, WHERE NAV2 GOAL FAILURE CAN ALSO COST A TASK. AHE-MRTA LEADS EVERY SCENARIO ON BOTH PLANES; EVERY PAIRED-WILCOXON COMPARISON AGAINST A BASELINE IS SIGNIFICANT EXCEPT ONE (ALL $p < 0.143$ EXCEPT AHE-BiG UNDER ROBOT FAILURE, WHICH IS A TIE). **BOLD** MARKS THE BEST PER COLUMN.

Method	Perfect nav (alloc. quality)			Stochastic nav (resil.)		
	RF	MS	DP	RF	MS	DP
AHE-MRTA*	0.994	0.727	0.840	0.384	0.263	0.349
BiG-MRTA	0.980	0.668	0.733	0.380	0.229	0.284
RoSTAM-EA	0.935	0.643	0.679	0.333	0.225	0.227
Cons-DBTA	0.978	0.642	0.729	0.364	0.217	0.258

left unserved: priority-weighted on-time completion runs from 0.647 to 1.000. Deadline violation is what separates the cells, and it tracks the scenario rather than the scale: 0.000–0.036 under robot failure, 0.116–0.232 under deadline pressure, and 0.215–0.353 under mixed stress. The active-robot Jain index is 0.983–0.994 at every scale, and decision latency is at most 0.215 ms, in the 10r/50t mixed-stress cell. During development, six 10r seeds allowed one task to execute on two robots, which inflated CR to 1.02. After adding the in-flight owner lock, we reran those seeds and the complete 10r campaign. CR is then exactly 1.000 in every cell.

Two things follow. Fitness rises with fleet size in both stressed scenarios (0.774 $\rightarrow$ 0.888 under deadline pressure and 0.647 $\rightarrow$ 0.782 under mixed stress, from 3 to 10 robots), which is what more robots buy against unchanged per-task budgets. And the plane separates the policies: at the primary scale AHE-MRTA leads every scenario—0.994, 0.727 and 0.840 against 0.980, 0.668 and 0.733 for the best baseline in each, BiG-MRTA in all three (Table V, left). Deterministic navigation therefore does not put the leading methods at a common ceiling; it removes navigation failure as a confound and leaves allocation and queue order to do the separating.

A. Separate resilience experiment: navigation-proxy

The next experiment simplifies navigation *cost* to a fixed-time edge in the robot–task graph. Navigation *outcomes* remain position-dependent and stochastic: a goal can fail and must be reattempted. This is not a navigation-independent setting. We use it as a navigation proxy for recovery after failed goals. Fitness is the priority-weighted fraction of tasks completed before their deadline. Because a failed goal must be recovered, fitness captures both allocation and resilience to navigation failure. With 500 seeds per scenario, AHE-MRTA leads every scenario on both planes (Table V), and eight of the nine paired comparisons against a baseline are significant, most of them overwhelmingly (p from 10^{-9} to 10^{-76}). The exception is a tie with BiG-MRTA under robot failure ($+0.004$, $p=0.14$).

At the primary scale (25 tasks and 5 robots; Table V, left), no method reaches the ceiling: with the deadline budgets the scenarios actually specify, on-time completion separates the four policies even when navigation cannot fail. AHE-MRTA leads at 0.994, 0.727, and 0.840 for robot failure, mixed stress, and deadline pressure, against 0.980, 0.668, and 0.733 for the best baseline in each. When navigation becomes stochastic (Table V, right) every method falls further, and the ordering is preserved: 0.384, 0.263, and 0.349 against 0.380, 0.229, and 0.284. The margin is smallest under robot failure, where BiG-MRTA’s commit-once discipline is well matched to the regime and the difference is not significant; it is largest under deadline pressure, which is the regime the overrides and the geodesic ETA both target.

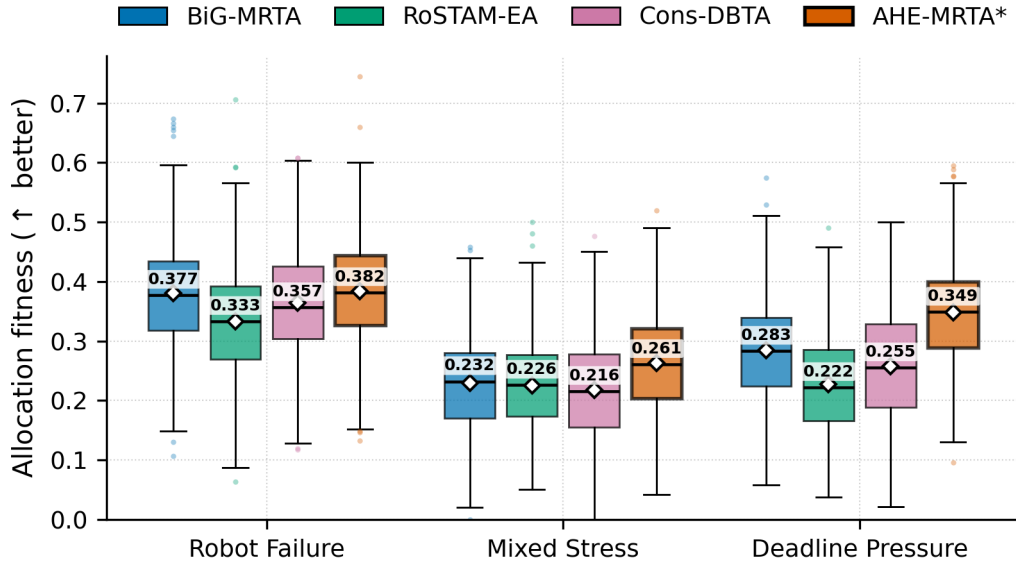


Fig. 4. Navigation-proxy fitness at the primary 5-robot/25-task scale (500 seeds per cell). Boxes show the interquartile range, centre lines the median (value annotated above each box), diamonds the mean, and whiskers 1.5 interquartile ranges; points are outliers. AHE-MRTA leads every scenario; the paired-Wilcoxon comparisons against each baseline are significant in eight of nine cases, the exception being a tie with BiG-MRTA under robot failure ($p=0.14$).

TABLE VI
SCALABILITY (STOCHASTIC NAVIGATION PROXY, GEODESIC EXECUTION ORACLE, 100 SEEDS, FIXED DENSITY 5 TASKS/ROBOT, MEAN OVER SCENARIOS). FITNESS $\uparrow$, LATENCY (MS) $\downarrow$. BEST PER SCALE **BOLD**. PHYSICAL GAZEBO VALIDATION: 3R, 5R, AND 10R CONFIRMED.

Method	3 robots		5 robots		10 robots	
	Fit.	Lat.	Fit.	Lat.	Fit.	Lat.
AHE-MRTA*	0.300	0.12	0.329	0.20	0.326	0.54
BiG-MRTA	0.267	0.02	0.293	0.05	0.298	0.17
RoSTAM-EA	0.249	1.93	0.261	3.48	0.245	7.00
Cons-DBTA	0.257	0.02	0.276	0.03	0.290	0.05

An earlier version of this evaluation reported these two planes as saturated and unable to separate the leading policies. That was an artefact of the simulator’s scenario parameters having drifted from the ones documented in Section IV-E: deadline budgets were drawn from $\mathcal{U}[200, 400]$ s where the design specifies $\mathcal{U}[36, 120]$ s under deadline pressure, and the halved budget and staggered arrivals of mixed stress were absent. Under budgets that generous every policy completes almost everything on time, so nothing could separate. Both planes now read their scenario definitions from the same module the Gazebo runner uses, and a parity test fails if the two ever diverge again. The proxy still does not isolate which mechanism produces the margin, and it does not predict a closed-loop Nav2 result: it charges no cost for congestion, replanning latency, or recovery time, all of which the physical stack does. Section VI reports the Gazebo estimates separately, and Section VII-E is where the mechanism question is settled. Figure 4 summarizes the proxy results.

B. Navigation-proxy scalability

We examine scalability in two settings. In the navigation proxy, the simplified arrival model permits a high-power sweep over robot count. We run four methods at $N \in \{3, 5, 10\}$ robots, with five tasks per robot ($M = 5N$), 100 seeds per cell, and all three scenarios. This produces 3,600 runs. We also evaluate closed-loop Gazebo execution at 3r, 5r, and 10r. Table VI and Figure 5 average proxy results over scenarios.

AHE-MRTA is highest on proxy fitness at every scale (Table VI), by 3.3, 3.6, and 2.7 pp over the best baseline at $N = 3$, 5, and 10; the runner-up is BiG-MRTA at all three. The margin does not grow with fleet size, so we read the result as a consistent advantage rather than a scaling one. Proxy decision latency remains about 0.54 ms at 10 robots. This is nearly four orders of magnitude below the 5 s replanning period and about 13 times lower than RoSTAM-EA (7.0 ms at $N = 10$). In the closed-loop Nav2 stack the same allocator decides in 0.5–3.4 ms, once the geodesic oracle is built at start-up rather than inside the first decision (Section VI).

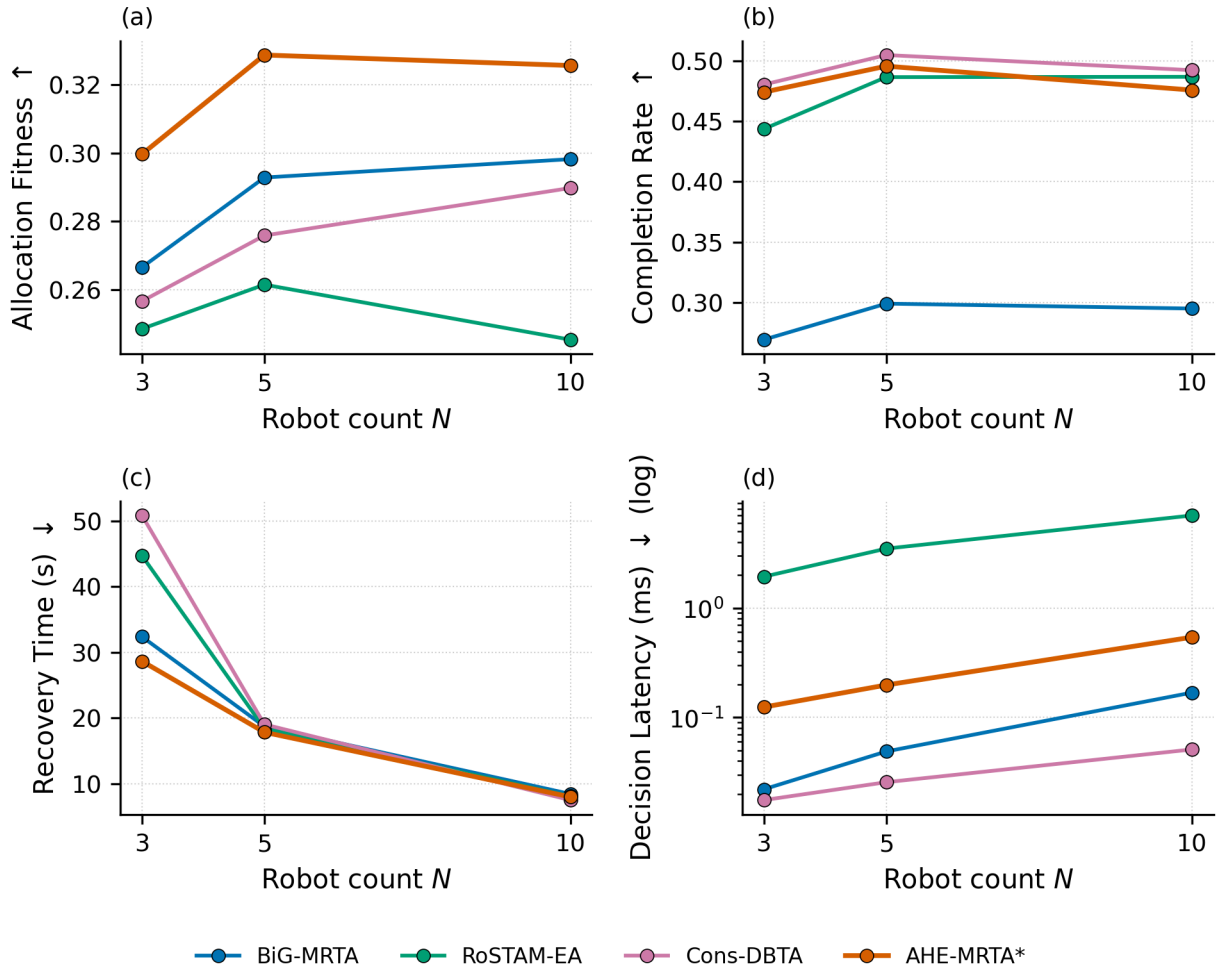


Fig. 5. Scalability by robot count in the stochastic navigation proxy. Values are means over three scenarios with 100 seeds per cell: (a) fitness, (b) completion rate, (c) recovery time, and (d) decision latency. AHE-MRTA is highest on fitness at every scale, ahead of the runner-up (BiG-MRTA) by 3.3, 3.6, and 2.7 pp at $N = 3, 5$, and 10; the margin does not grow with fleet size.

The 5-robot Gazebo study contains 240 runs and the 10-robot study 60. Their per-scenario results appear with the 3-robot benchmark in Section VI. They are not a direct confirmation of the proxy because they use a different noise model and, at 10 robots, far fewer seeds.

VI. GAZEBO BENCHMARK RESULTS

We report the closed-loop Gazebo benchmark. Methods use identical Nav2 parameters, world, and timing budgets; only the allocation policy differs.¹ At 3r, three density levels (9, 15, and 24 tasks) with five seeds per method–scenario combination give 180 runs. The 5r study adds 240 (20 seeds per method–scenario combination) and the 10r study 60, for 480 runs in total. We use Bonferroni-corrected Mann–Whitney U tests as described in Section IV-H.

Figure 6 shows the occupancy map and representative trajectories from one *mixed_stress* instance (5 robots, 25 tasks, seed 1). We reconstruct trajectories from logged ground-truth poses and overlay them on the arena and sampled waypoints. In this example, AHE-MRTA and Consensus-DBTA distribute robots across the arena and cover every waypoint. Commit-once BiG-MRTA leaves a larger area uncovered. This is a qualitative illustration, not an aggregate comparison.

Figure 7 shows a mid-run snapshot of the 10-robot configuration under AHE-MRTA. The snapshot comes from an illustrative 10-robot / 50-task run with staggered arrivals and no injected failure; it is outside the three benchmark scenarios and contributes to no reported number. The Gazebo view shows robots travelling to assigned waypoints. The concurrent RViz view shows the occupancy map, each robot’s Nav2 global plan, and its ground-truth pose. The snapshot illustrates work-conserving dispatch across both lanes; it is not a quantitative result.

¹Tables and figures denote AHE-MRTA as AHE-MRTA* where it is evaluated with its geodesic-ETA cost oracle and terminal load repair (Sec. III-D).

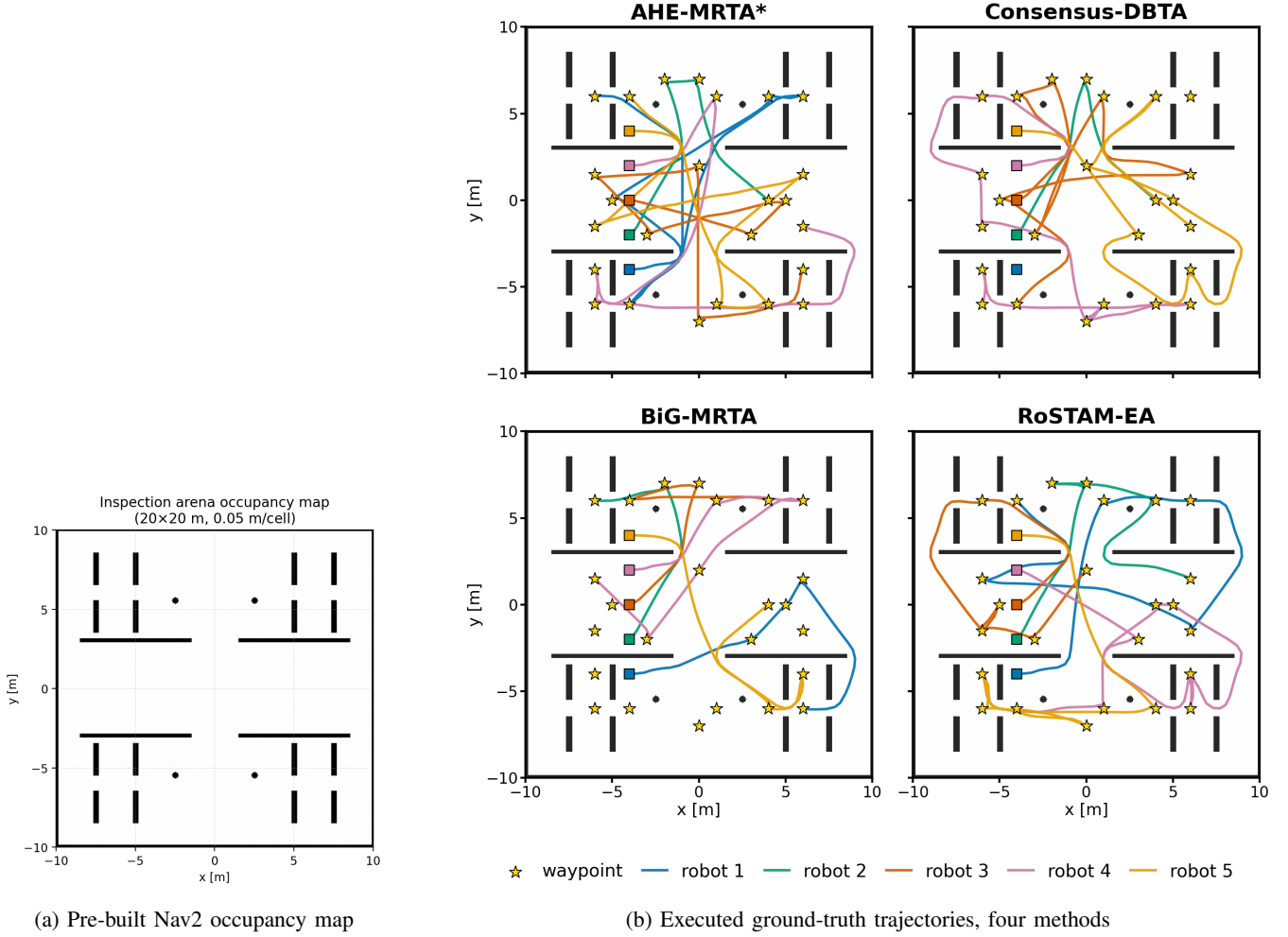


Fig. 6. Experimental map and representative trajectories. (a) The static Nav2 occupancy map (`obstacle_map`) is 20×20 m with 0.05 m cells and origin $[-10, -10]$. It contains divider walls, square pillars, and four cylindrical obstacles. (b) Paths from one *mixed_stress* run (5 robots, 25 tasks, seed 1) for AHE-MRTA, Consensus-DBTA, BiG-MRTA, and RoSTAM-EA, in reading order from top left. These are the same four runs that enter the *5r/25t mixed_stress* cells of Table VII. Squares are spawn positions, gold stars are sampled waypoints, and coloured curves are paths reconstructed from ground-truth poses; the timeseries logs each pose in its robot’s odom frame, so every trace is shifted by that robot’s spawn point. A square without a curve is a robot that never left its spawn, as Consensus-DBTA’s robot 1 does here. This single-run figure is qualitative; it does not establish an aggregate method ranking.

a) The deadline-pressure result.: The clearest closed-loop separation is under *deadline_pressure*, and it is the same at both physical scales. Effective on-time completion, $\text{CR} \times (1 - \text{DVR})$, is 0.650 for AHE-MRTA* at 5 robots against 0.578 for the strongest baseline, and 0.764 against 0.524 at 10 robots. The margin comes from deadline compliance, not from throughput. Three of the four methods complete essentially every task at 5 robots, so the separation lies entirely in DVR: AHE-MRTA* holds it at 0.350, while the strongest baseline by the same measure, Consensus-DBTA, records 0.422. That is a 17% relative reduction. At 10 robots the same comparison is 0.220 against 0.428, a 49% reduction, and the baselines reach their completion rates only by tolerating those violations. This is the pattern the method was built for: the deadline override and the geodesic ETA both act on time slack, and this is the regime where slack is scarce.

At the primary scale these are corrected results, not point estimates: with $n=20$ per cell, 20 of 63 cross-method comparisons survive Bonferroni correction, 15 of them in AHE-MRTA’s favour. The five that go the other way are two distinct trade-offs, both itemised later in this section: decision latency against commit-once BiG-MRTA in all three scenarios and against Consensus-DBTA under mixed stress, and average task delay against Consensus-DBTA under robot failure. The 10-robot cells still carry $n=5$ and remain descriptive.

b) Descriptive Gazebo patterns.: Table VII reports the *robot_failure* and *mixed_stress* cells at the primary scale and Table VIII the *deadline_pressure* cell; Figure 8 shows the same scale across all six primary metrics. At this scale AHE-MRTA completes every task in every scenario ($\text{CR}=1.000$); at 10 robots it does not, reaching 0.976–0.988. BiG-MRTA has lower point estimates under *deadline_pressure* (0.664) and *mixed_stress* (0.656).

The four-method comparison cannot attribute a Gazebo difference to the selector because methods differ in several components. The matched fixed-paradigm and no-override ablation of Section VII-E is what separates those factors, and it assigns

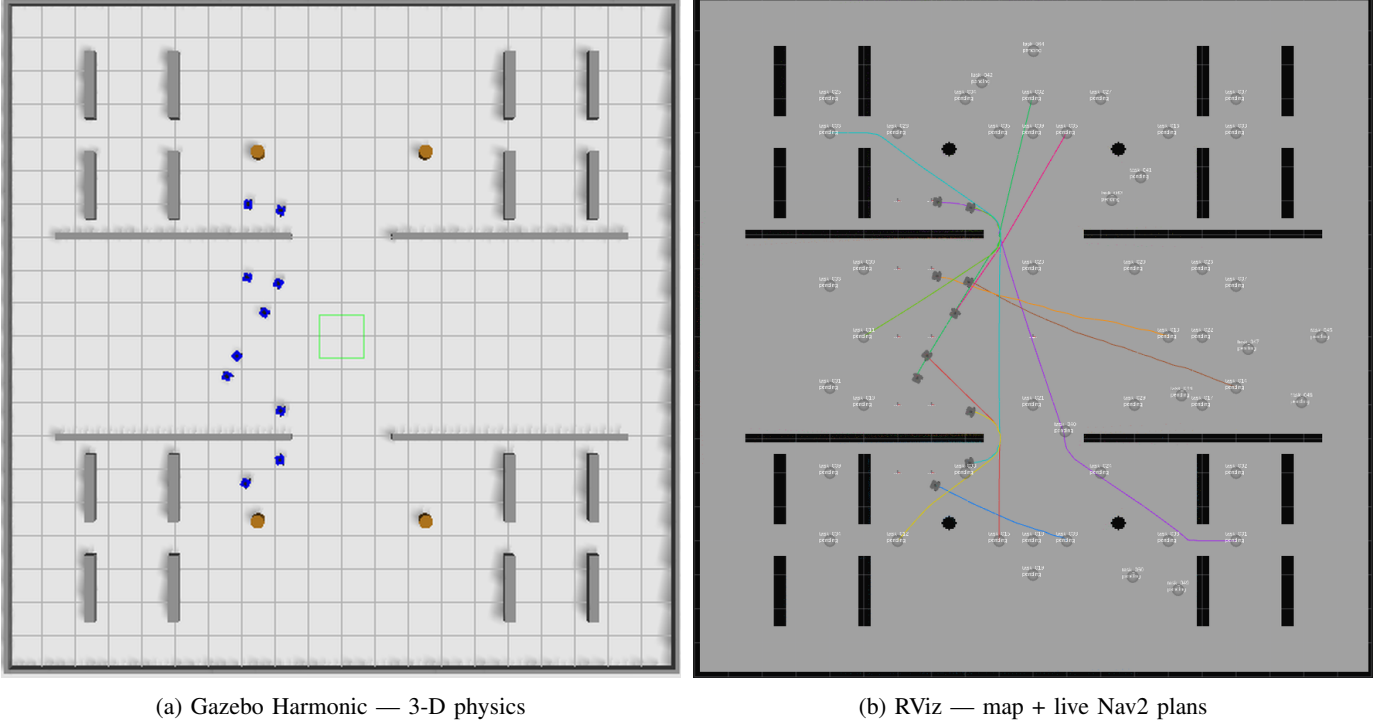


Fig. 7. Representative live snapshot of an illustrative 10-robot / 50-task AHE-MRTA run with staggered task arrivals, outside the three benchmark scenarios. Both panels are captured from the same paused simulation instant of the same run, so robot positions correspond one-to-one. (a) Gazebo Harmonic (top-down view) shows the ten TurtleBot3 WafflePi robots travelling in the 20×20 m arena. (b) RViz shows the occupancy map, current Nav2 global plan for each robot, and ground-truth-anchored poses. The image illustrates the execution state of one run; it is not comparative evidence.

TABLE VII

MAIN COMPARISON AT THE PRIMARY 5-ROBOT / 25-TASK SCALE; $n=20$ RUNS (SEEDS) PER CELL. BEST VALUE PER COLUMN (WITHIN SCENARIO) IN **BOLD**. SIGNIFICANCE VS AHE-MRTA*: * $p<0.05$, ** $p<0.01$, *** $p<0.001$ (MANN-WHITNEY U, BONFERRONI-CORRECTED WITHIN EACH SCENARIO FAMILY OF 21 TESTS).

Method	CR $\uparrow$	Delay $\downarrow$ (s)	RecT $\downarrow$ (s)	Preempt $\downarrow$	Churn $\downarrow$
<i>Scenario: robot_failure</i>					
AHE-MRTA*	1.000 $\pm$ 0.000	99.7 $\pm$ 7.6	83.7 $\pm$ 43.8	0.00 $\pm$ 0.00	0.006 $\pm$ 0.015
BiG-MRTA	0.956*** $\pm$ 0.036	104.4 $\pm$ 30.1	73.1 $\pm$ 33.9	0.00 $\pm$ 0.00	0.004 $\pm$ 0.013
RoSTAM-EA	1.000 $\pm$ 0.000	92.6 $\pm$ 10.8	74.5 $\pm$ 35.8	0.00 $\pm$ 0.00	0.012 $\pm$ 0.019
Cons-DBTA	1.000 $\pm$ 0.000	86.0** $\pm$ 11.2	117.9 $\pm$ 29.4	0.00 $\pm$ 0.00	0.006 $\pm$ 0.015
<i>Scenario: mixed_stress</i>					
AHE-MRTA*	1.000 $\pm$ 0.000	65.3 $\pm$ 9.1	41.9 $\pm$ 17.6	0.00 $\pm$ 0.00	0.014 $\pm$ 0.020
BiG-MRTA	0.656*** $\pm$ 0.060	311.2*** $\pm$ 46.1	22.8 $\pm$ 23.7	0.00 $\pm$ 0.00	0.018 $\pm$ 0.029
RoSTAM-EA	1.000 $\pm$ 0.000	61.9 $\pm$ 10.1	85.3 $\pm$ 56.0	0.00 $\pm$ 0.00	0.002 $\pm$ 0.009
Cons-DBTA	0.998 $\pm$ 0.009	72.8 $\pm$ 16.4	81.5 $\pm$ 82.3	0.00 $\pm$ 0.00	0.032 $\pm$ 0.098

the pattern to the portfolio rather than to the override cascade.

c) *Costs of AHE.*: Reactive behavior can increase execution costs, and at the primary scale two of them are now measurable rather than merely suggestive. Under *robot_failure*, AHE's average delay is 99.7 s against Consensus-DBTA's 86.0 s—a corrected-significant gap ($p_{\text{adj}}=0.002$, $\delta=0.72$) and one of the five comparisons that go against AHE-MRTA. RoSTAM-EA (92.6 s) and commit-once BiG-MRTA (104.4 s) sit either side of it and neither separates. Its 83.7 s recovery time is lower than Consensus-DBTA's 117.9 s and higher than RoSTAM-EA's 74.5 s and BiG-MRTA's 73.1 s, but recovery time separates from no baseline (Figure 10); the cell is confounded by Nav2 motion in any case. Under *mixed_stress* and *deadline_pressure*, BiG-MRTA has the largest delay (311–327 s) because its completed tasks finish very late; this is not a like-for-like comparison when it abandons other tasks. On redispatch churn, AHE is in the best band: at most 0.014 redispaches per task and zero execution preemptions. We count executed reassignments here; the `allocation_instability` counter logged by the runner counts allocator invocations instead, and reports values an order of magnitude larger for every method (e.g., 1.02 for AHE and 5.29 for BiG-MRTA in this cell), so the two must not be compared. Decision latency is the second measurable cost, and the direction depends on the comparison. The geodesic ETA oracle depends only on the static map and the task pool, both known before the run, so it is built once at start-up instead of inside the first timed decision; AHE-MRTA then

TABLE VIII

DEADLINE SCENARIO RESULTS (DEADLINE_PRESSURE) AT THE PRIMARY 5-ROBOT / 25-TASK SCALE; $n=20$ RUNS (SEEDS) PER CELL. DVR: DEADLINE VIOLATION RATE. BEST VALUE PER COLUMN IN **BOLD**. SIGNIFICANCE VS AHE-MRTA* (BONFERRONI-CORRECTED MANN-WHITNEY U).

Method	CR $\uparrow$	Delay $\downarrow$ (s)	DVR $\downarrow$	Preempt $\downarrow$
AHE-MRTA*	1.000 $\pm$ 0.000	78.5 $\pm$ 7.7	0.350 $\pm$ 0.090	0.00 $\pm$ 0.00
BiG-MRTA	0.664*** $\pm$ 0.059	327.0*** $\pm$ 46.5	0.402 $\pm$ 0.073	0.00 $\pm$ 0.00
RoSTAM-EA	1.000 $\pm$ 0.000	77.8 $\pm$ 7.3	0.498*** $\pm$ 0.073	0.00 $\pm$ 0.00
Cons-DBTA	1.000 $\pm$ 0.000	72.6 $\pm$ 9.3	0.422 $\pm$ 0.107	0.00 $\pm$ 0.00

TABLE IX

EFFICIENCY METRICS (GAZEBO, 3-ROBOT SCALE; DENSITIES 9/15/24 POOLED, $n=15$ PER CELL). WLbal: ALL-ROBOT JAIN WORKLOAD BALANCE; LAT: MEAN DECISION LATENCY; MSGS: ALLOCATION MESSAGES; DIST: TOTAL TRAVEL DISTANCE. BEST VALUE PER COLUMN (WITHIN SCENARIO) IN **BOLD**. AHE-MRTA* LATENCY IS THE WARM-UP RE-MEASUREMENT AT THE SAME $n=15$; ALL OTHER ENTRIES COME FROM THE ORIGINAL CAMPAIGN.

Method	WLbal $\uparrow$	Lat $\downarrow$ (ms)	Msgs $\downarrow$	Dist $\downarrow$ (m)
<i>Scenario: robot_failure</i>				
AHE-MRTA*	0.853	0.49	32	108.0
BiG-MRTA	0.949	0.13	131	72.8
RoSTAM-EA	0.947	40.24	2	84.4
Cons-DBTA	0.891	0.29	12	90.9
<i>Scenario: mixed_stress</i>				
AHE-MRTA*	0.857	0.47	35	113.1
BiG-MRTA	0.928	0.08	177	57.4
RoSTAM-EA	0.931	30.84	4	93.7
Cons-DBTA	0.875	0.17	10	104.0
<i>Scenario: deadline_pressure</i>				
AHE-MRTA*	0.986	0.72	25	99.2
BiG-MRTA	0.965	0.09	180	43.9
RoSTAM-EA	0.978	48.86	1	76.7
Cons-DBTA	0.997	0.51	6	86.4

decides in 0.81–1.33 ms at 5 robots and 1.9–3.4 ms at 10. That is corrected-significantly slower than commit-once BiG-MRTA (0.14–0.24 ms at 5 robots) in all three scenarios, and slower than Consensus-DBTA under *mixed_stress*, while remaining faster than Consensus-DBTA under the other two and one to two orders of magnitude faster than RoSTAM-EA. Three of the four stay far inside the 50 ms budget, so among them the ordering is a property of the allocators rather than a constraint on deployment; RoSTAM-EA does not (49–95 ms at this scale), and for it the ordering is a deployment constraint as well.

d) *Efficiency.*: Table IX gives secondary efficiency metrics at the 3-robot scale, where the three densities pool to $n=15$ per cell. There AHE-MRTA takes 0.47–0.72 ms per decision, against 0.08–0.13 ms for BiG-MRTA, 0.17–0.51 ms for Consensus-DBTA, and 30.8–48.9 ms for RoSTAM-EA. The corresponding figures at the primary 5-robot scale are 0.81–1.33 ms for AHE-MRTA, 0.14–0.24 ms for BiG-MRTA, 0.45–2.43 ms for Consensus-DBTA, and 49–95 ms for RoSTAM-EA. AHE-MRTA therefore stays within the 50 ms budget at every scale; RoSTAM-EA does not.² Consensus methods have nearly perfect workload balance and few messages because they commit assignments. AHE trades some of this balance for reactive changes and the reported deadline outcomes: at 3 robots its all-robot Jain balance is 0.85–0.99, against 0.93–0.96 for commit-once BiG-MRTA and 0.88–1.00 for Consensus-DBTA, and its routes are longer (99–113 m versus 44–104 m) because redispached tasks are re-approached from a different robot.

e) *Effect sizes.*: Table X reports Cliff’s δ alongside the corrected tests, so that the size of a difference can be read independently of whether it clears the threshold. Its sign is normalized so $\delta > 0$ favors AHE. Completion effects versus BiG-MRTA are large: +1.00 under *mixed_stress* and *deadline_pressure*, and +0.70 under *robot_failure*. Deadline-compliance effects are uniformly large under *robot_failure* (+0.83 to +0.86 against all three baselines, all corrected-significant); under *deadline_pressure* only the RoSTAM-EA comparison is both large and significant (+0.81), with BiG-MRTA at +0.34 and Consensus-DBTA at +0.39. The largest unfavorable effect is recovery time versus commit-once BiG-MRTA under *mixed_stress* (−0.50), which does not survive correction; recovery against reactive Consensus-DBTA runs the other way (+0.45 under

²At the primary 5-robot scale the campaign runs were themselves executed with the geodesic oracle built at start-up, so the reported AHE-MRTA latency is the campaign’s own. The 3-robot and 10-robot AHE-MRTA latency figures instead come from a 60-run re-measurement that matches those cells of the original campaign one for one (four scale-density cells $\times$ three scenarios $\times$ five seeds); every other AHE-MRTA quantity and all baseline quantities come from the campaign itself. The change moves map-derived set-up out of the timed path and leaves allocation decisions bit-identical, so no other metric is affected: in a paired control at 5r/25t the unmodified arm reproduced the pre-warmup campaign’s latency (15.4 versus 15.9 ms) while every non-latency difference stayed inside ordinary Gazebo run-to-run variation.

TABLE X
EFFECT SIZES (CLIFF'S δ) OF AHE-MRTA* VS EACH BASELINE ON PRIMARY METRICS. $\delta > 0$ FAVORS AHE-MRTA*. * $p < 0.05$ (MANN-WHITNEY U, BONFERRONI-CORRECTED WITHIN SCENARIO FAMILY).

Metric	vs	BiG	RoSTAM	Cons-DBTA
<i>Scenario: robot_failure</i>				
CR	δ	+0.70***	+0.00	+0.00
DVR	δ	+0.83***	+0.86***	+0.85***
RecT	δ	-0.08	-0.09	+0.45
Churn	δ	-0.04	+0.15	-0.00
<i>Scenario: mixed_stress</i>				
CR	δ	+1.00***	+0.00	+0.05
DVR	δ	-0.03	+0.47	+0.65**
RecT	δ	-0.50	+0.49	+0.18
Churn	δ	+0.06	-0.30	-0.03
<i>Scenario: deadline_pressure</i>				
CR	δ	+1.00***	+0.00	+0.00
DVR	δ	+0.34	+0.81***	+0.39
RecT	δ	-0.00	-0.00	-0.00
Churn	δ	-0.00	-0.00	-0.00

Positive δ = AHE-MRTA* better; magnitude $|\delta| > 0.47$ = large.

robot_failure). On corrected redispatch churn no comparison is larger than $|\delta| = 0.30$ in either direction. Read with the tests, these sizes say the deadline-compliance advantage is the substantial one and the execution-cost differences are mostly small.

f) *Statistical significance.*: At the primary 5-robot scale ($n=20$ per cell), 20 of 63 cross-method comparisons survive Bonferroni correction, 15 in AHE-MRTA's favour and 5 against it; 28 of 63 reach raw $p < 0.05$. The 15 favourable results are completion rate and deadline violation rate against the baselines that lose on each, plus decision latency against RoSTAM-EA and Consensus-DBTA; the 5 unfavourable ones are decision latency against commit-once BiG-MRTA in all three scenarios, decision latency against Consensus-DBTA under *mixed_stress*, and average task delay against Consensus-DBTA under *robot_failure*. At 3 robots ($n=15$ per cell), 14 of 63 comparisons survive correction, 8 of them in AHE-MRTA's favour. At 10 robots the five seeds per cell cannot support correction at all—the smallest attainable two-sided Mann-Whitney value is about 0.008, above the corrected threshold of about 0.0024—so none survive and those cells stay descriptive. The 500-seed navigation-proxy study (Section V) remains the highest-powered cross-method test; the Gazebo benchmark now carries corrected inference of its own at the primary scale.

g) *Descriptive Gazebo results at 5 and 10 robots.*: Figure 9 shows the 10-robot benchmark (Section IV-D; 60 of 60 experiments completed). AHE-MRTA attains the highest—or tied-highest—completion rate in every scenario (CR=0.976–0.988). Its clearest lead is under *deadline_pressure*: it reaches CR=0.980 at DVR=0.220, giving the highest effective on-time completion ($CR \times (1 - DVR) \approx 0.76$) far ahead of Consensus-DBTA (≈ 0.52) and RoSTAM-EA (≈ 0.48), which reach CR=0.916/0.964 only by tolerating DVR=0.428/0.504. Under *mixed_stress* AHE posts the top completion rate (CR=0.988) but Consensus-DBTA edges it on effective on-time completion (0.650 versus 0.641), both well above RoSTAM-EA (0.529). Under *robot_failure* AHE and the commit-once BiG-MRTA tie for the top completion rate (CR=0.976), with BiG-MRTA slightly ahead on effective on-time completion (0.936 versus 0.915); the task-abandoning BiG-MRTA records its low DVR only where it completes far fewer tasks in the other two scenarios (CR=0.696–0.724). At the 5-robot scale AHE reaches full completion in every scenario (CR=1.000) and leads effective on-time completion throughout: *robot_failure* 0.986 (DVR=0.014) vs ≤ 0.890 , *deadline_pressure* 0.650 (DVR=0.350) vs ≤ 0.578 , and *mixed_stress* 0.616 vs ≤ 0.542 ; BiG-MRTA abandons tasks under load (CR=0.656/0.664). Decision latency is 0.81–1.33 ms at 5 robots and 1.9–3.4 ms at 10—within the spread of the reactive baselines and 20–82 $\times$ below RoSTAM-EA, which grows to 48–107 ms at 10 robots—and far within the real-time budget at both scales.

At the 3-robot scale, Figure 11 traces the mechanism behind these aggregates on *robot_failure*: the dominance state, the seed-averaged completion curve across the injection, and the per-run spread of recovery time and delay. Re-dispatch is not plotted because it does not vary at this scale: the median is zero for every method, and the entire non-zero mass is 2 of 15 runs for AHE-MRTA and for Consensus-DBTA, 4 of 15 for BiG-MRTA, and none for RoSTAM-EA, with the largest single run at 0.11 re-dispatches per task for AHE-MRTA against 0.30 for Consensus-DBTA. Figure 12 gives the corresponding cumulative-completion curves for *robot_failure* and *mixed_stress*.

VII. DISCUSSION

A. Why does ideal-fitness not transfer linearly to Gazebo?

The deterministic allocation-only simulator (100 seeds per cell) removes Nav2 outcome variance such as local-planner stalls, costmap rebuilds, and passing manoeuvres in narrow corridors. It therefore realizes each accepted route according to the shared

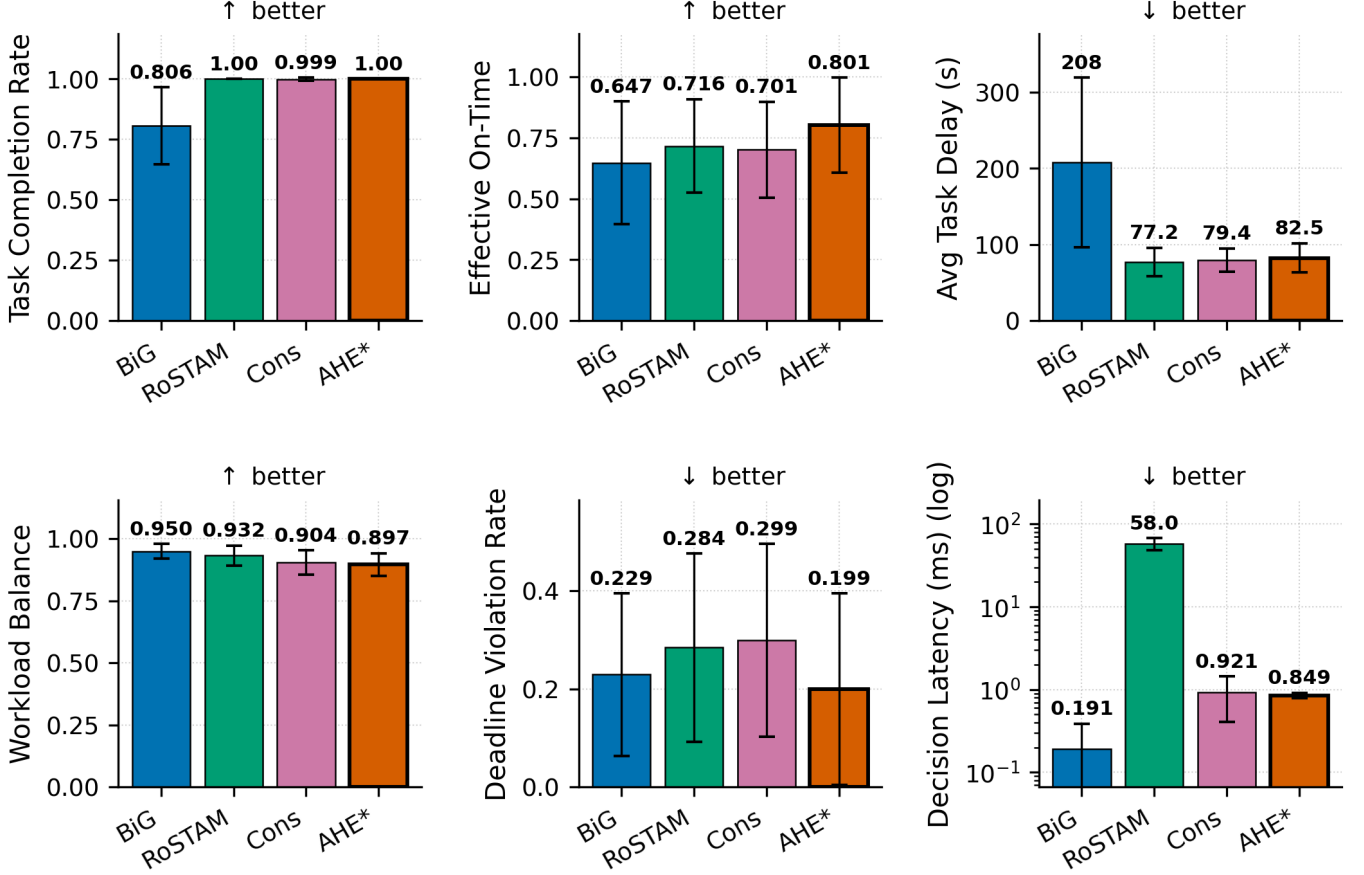


Fig. 8. Multi-metric results at the primary 5-robot/25-task scale (Gazebo Harmonic; *robot_failure* and *mixed_stress* pooled; $n=20$ seeds per cell; mean $\pm$ standard deviation). Panels show completion rate, effective on-time completion $CR \times (1 - DVR)$, average delay, workload balance, deadline violation rate, and log decision latency. Pooling the two scenarios is for display only; the corrected tests are run within each scenario and are reported in the text.

geodesic oracle. In Gazebo, two execution effects can reduce an allocation advantage:

- 1) **Switching has an execution cost.** A paradigm change can reassign a queued task. The new robot must plan and travel to it. A commit-once method avoids this replanning overhead.
- 2) **Geometry dominates makespan.** After Nav2 dispatches a robot, path length and local motion depend strongly on the map. Extra replanning can increase the longest remaining route.

The Gazebo data is consistent with these effects, but they show up in route cost rather than in the primary outcomes. AHE-MRTA keeps its completion and deadline advantage in the closed loop: it is the completion leader or co-leader in every cell, and it cuts DVR by 17% (5r) and 49% (10r) against the strongest baseline. It pays for the extra replanning elsewhere—in total travel distance (99–113 m versus 44–104 m at 3 robots) and, at the smaller scales, in makespan under *mixed_stress* (243 s versus 182–219 s for the two other reactive baselines at 3 robots; commit-once BiG-MRTA runs to the 900 s horizon in this cell). Redispatch churn is *not* where the cost appears at the primary scale: at 5 robots executed reassignments stay at or below 0.014 per task, the best band of the four methods. At 10 robots they rise to 0.06–0.08 per task, level with Consensus-DBTA (0.04–0.13) and above commit-once BiG-MRTA (0.005–0.03).

B. Observed operating patterns and attribution boundary

The following per-cell patterns describe the evaluated full configuration. They are not causal estimates of paradigm switching: the four-method comparison varies several components at once. The matched ablation of Section VII-E is what isolates the selector.

- **Deadline-compliance pattern.** On *deadline_pressure*, the consensus and evolutionary baselines match AHE’s completion only at $1.2\text{--}2.3\times$ its deadline-violation rate (5r: AHE DVR=0.350 vs Cons-DBTA’s 0.422 and RoSTAM-EA’s 0.498; 10r: AHE 0.220 vs 0.428 and 0.504). AHE-MRTA has the largest effective on-time completion at both scales (0.650 at 5r, 0.764 at 10r). The four-method comparison supports this operating pattern, but not a selector-only explanation.
- **Failure-recovery pattern.** Under *robot_failure* AHE combines full completion at 5 robots (CR=1.000, DVR=0.014, effective on-time 0.986 vs ≤ 0.890) with tied-highest completion at 10 robots (CR=0.976, level with BiG-MRTA on

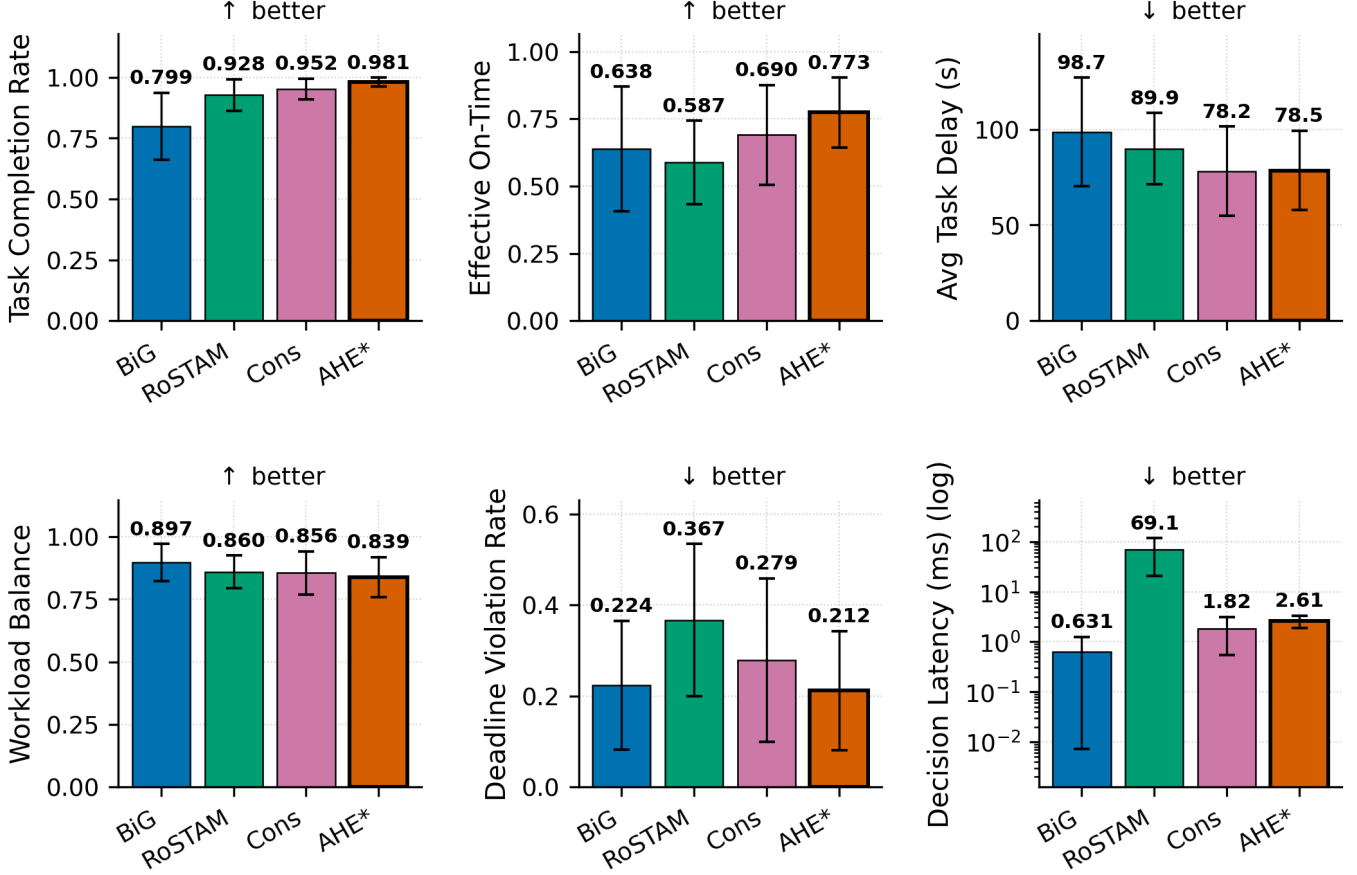


Fig. 9. Multi-metric results at the 10-robot scale (60 runs: four methods, three scenarios, and five seeds; the three scenarios are pooled, so each bar is $n=15$; mean $\pm$ standard deviation; decision latency on a log scale). Pooled over scenarios, AHE-MRTA leads on completion (0.981 versus 0.799–0.952), on effective on-time completion $CR \times (1 - DVR)$ (0.773 versus 0.587–0.690), and is lowest on deadline violation (0.212 versus 0.224–0.367), at 2.6 ms mean decision latency against 0.6 ms for commit-once BiG-MRTA, 1.8 ms for Consensus-DBTA, and 69.1 ms for RoSTAM-EA. Pooling hides where the deadline margin actually comes from: it is concentrated in *deadline_pressure* (DVR 0.220 versus 0.296–0.504), which the text reports per scenario. The $n=5$ per-scenario cross-method cells are descriptive and do not establish a corrected ranking.

effective on-time). The implementation invokes an orphan-rescue behaviour after a failure, but the present four-method comparison does not identify its marginal causal effect.

- **Completion stops separating methods at the small scale, and the residual edge is paid for in makespan.** Under uniform deadline pressure at 3 robots the EDF-style specialists already reach full completion in Gazebo (RoSTAM-EA, Cons-DBTA and AHE-MRTA all at $CR=1.000$; only the task-abandoning BiG-MRTA trails at 0.633), so completion carries no information at this scale and the remaining separation is on deadline compliance alone (effective on-time 0.69 for AHE-MRTA versus 0.60–0.66 for the specialists). That margin is bought with a longer makespan (179 s versus 146–159 s), higher delay (80 s versus 70–77 s), and longer routes (99 m versus 77–86 m). The navigation proxy at the same scale likewise puts AHE-MRTA within noise of BiG-MRTA on strict on-time fitness (0.420 versus 0.417, Table VI). At 10 robots, however, AHE keeps DVR at 0.220 against the specialists’ 0.428–0.504 and attains the highest effective on-time completion (Sec. VI).

C. Why the navigation-proxy margins stay narrow

AHE-MRTA leads every scenario on the stochastic navigation-proxy fitness (priority-weighted on-time completion), but by margins of 0.004–0.065 rather than by a wide separation, and all four methods score far below the perfect-navigation plane (Table V). Two mechanisms explain both facts. First, the ceiling on this plane is set by *stochastic, position-dependent navigation failure*, not by assignment quality: the same policies score 0.642–0.994 when navigation cannot fail and 0.217–0.384 when it can, so most of the loss is failure and re-attempt, which no allocator can price away. The binding constraint is therefore resilience to navigation failure—a capability AHE-MRTA and Consensus-DBTA both implement by re-dispatching failed and orphaned tasks—which compresses the field. It compresses it most under robot failure, where commit-once BiG-MRTA is also well matched to the regime and finishes within 0.004 of AHE-MRTA, the one comparison on this plane that does not separate; the margins widen under mixed stress and deadline pressure (+0.034 and +0.065 over the best baseline), where

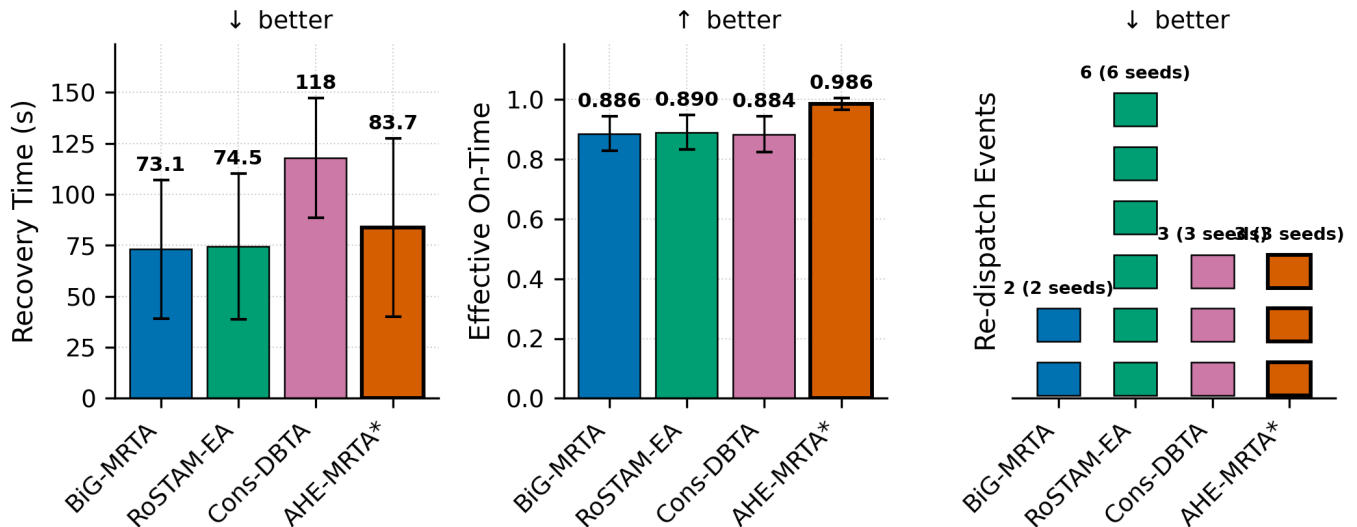


Fig. 10. Failure-recovery behaviour under *robot_failure* at the primary 5-robot / 25-task scale ($n=20$ seeds): recovery time and effective on-time completion, $CR \times (1 - DVR)$ (mean $\pm$ std bars, whiskers clipped at zero, since neither quantity can be negative), and re-dispatch events as a unit bar in which every square is a single event and squares from different seeds are separated by a gap. This keeps the rare, discrete nature of the metric visible: 66 of the 80 runs have zero re-dispatches, and each of the remaining 14 has exactly one, spread over distinct seeds—three for AHE-MRTA, two for BiG-MRTA, three for Consensus-DBTA, and six for RoSTAM-EA (execution preemptions are zero for every method and are omitted). No recovery-time comparison survives correction; the closest is against Consensus-DBTA, where AHE-MRTA’s 83.7 s beats 117.9 s at raw $p=0.017$ but corrected $p=0.35$. Raw completion rate is omitted because it does not discriminate here—AHE-MRTA, RoSTAM-EA and Consensus-DBTA are all at 1.000—whereas on effective on-time completion AHE-MRTA leads at 0.986 against 0.884–0.890, i.e. it converts a comparable recovery time into fewer late completions.

TABLE XI

SELECTOR ABLATION (STOCHASTIC NAVIGATION PROXY, GEODESIC EXECUTION ORACLE, 5R/25T, 100 SEEDS). ON THE CORRECTED SCENARIO PARAMETERS THE VARIANTS SPAN 0.101 IN MEAN FITNESS. REMOVING THE CONTEXT OVERRIDES LEAVES THE SELECTOR ESSENTIALLY UNCHANGED, AND FIXED SPATIAL-GREEDY IS AHEAD OF IT ON THIS PLANE—THE CLOSED-LOOP ABLATION (SECTION VII-E) TESTS BOTH OBSERVATIONS.

Variant	Rob. fail.	Deadline	Mix. stress	Mean
Full EDPS	0.383	0.348	0.255	0.329
fixed: spatial	0.421	0.358	0.265	0.348
fixed: priority	0.357	0.298	0.274	0.310
fixed: edf	0.333	0.296	0.276	0.302
fixed: commit	0.344	0.186	0.210	0.247
fixed: orphan	0.318	0.259	0.254	0.277
no-override	0.384	0.331	0.254	0.323
no-recovery-boost	0.382	0.348	0.255	0.328
Full selector mean 0.329; best variant 0.348.				

abandoning or mis-ordering a task is not free. Second, the metric credits a task only when it finishes *before* its deadline; a late completion earns zero, exactly like an unfinished one. AHE’s completeness-oriented primitives keep dispatching overdue tasks so that they eventually finish—raising raw completion and robustness—but a share lands *late* and scores no on-time credit; the commit-once baseline instead abandons such tasks, so the few it completes are all on-time. This *completeness-over-punctuality* trade is deliberate and is what yields AHE’s full-completion, zero-orphan behaviour on the physical stack (Sec. VI). We further probed whether a deadline-feasibility-aware execution ordering could widen the proxy margin: in isolation it lifts deadline fitness by ≈ 1.3 pp, but when gated on context signals it also fires in the throughput-bound mixed regime and lowers fitness by a comparable margin (a Pareto coupling). We therefore retain the simpler completeness-oriented policy, whose leading fitness and physical-stack advantages (Sec. VI) are the stronger operating point.

D. Ablation: what does the switching machinery add?

To isolate the contribution of online paradigm switching, we replace the selector with each single *fixed* paradigm, and separately disable the context overrides (*no-override*: pure $\arg \max_i D_i$) and the recovery boost (*no-recovery-boost*: $\delta=0$), holding everything else fixed (5r/25t, 100 seeds, stochastic navigation proxy with the geodesic execution oracle).

The variants span 0.101 in mean fitness, and two readings stand out. First, removing the context overrides is nearly free on this plane (0.323 against 0.329), as is removing the recovery boost (0.328). Second, one fixed paradigm—spatial-greedy—is

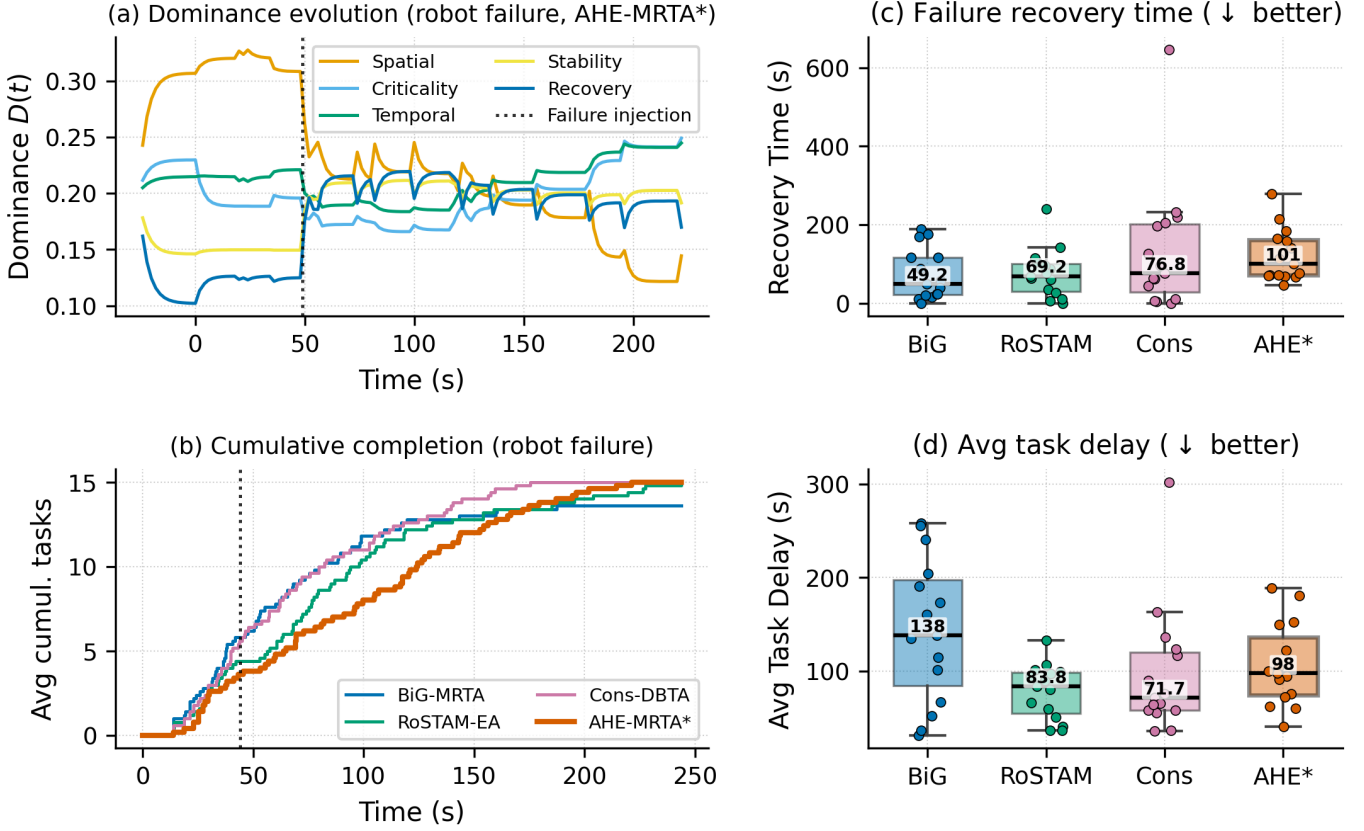


Fig. 11. Interpretable adaptation under *robot failure* at the 3-robot scale; panels (a)–(b) use the 15-task density and panels (c)–(d) pool the three densities. Time is measured from each run’s first task activation, so the injection markers in (a) and (b) sit at the scenario’s 45 ± 5 s offset. **(a)** Hormone dominance of a representative AHE-MRTA run; the spatial hormone leads the pre-failure phase ($D \approx 0.31$), and at the injection (dotted line, $t \approx 49$ s) it collapses while the recovery and stability hormones rise ($0.12 \rightarrow 0.20$ and $0.15 \rightarrow 0.20$). No single hormone then regains the pre-failure margin: the spatial channel keeps a reduced lead for about a minute, after which the temporal and criticality channels lead, and the dispatched paradigm alternates between temporal, spatial, and recovery passes. **(b)** Cumulative task completion (seed-averaged; dotted line: median injection $t \approx 44$ s): AHE-MRTA continues completing tasks after the failure and reaches the full task set, though it is the last of the three reactive methods to do so. **(c–d)** Recovery time and average task delay as box plots (box: IQR; line: median; whiskers: $1.5 \times \text{IQR}$; points: individual runs, $n=15$ per method): AHE’s reactivity shows up as the highest recovery-time median (101 s versus 49–77 s), while its delay median (98 s) sits between the consensus baselines (72–84 s) and the task-abandoning BiG-MRTA (138 s).

ahead of the full selector (0.348), while the other four fixed paradigms are behind it by 0.02–0.08. The proxy therefore suggests both that the override layer is inert and that the selector may be beaten by the single behaviour its fallback returns most often.

Neither reading transfers automatically: this plane charges nothing for congestion, replanning latency, or recovery time, which is where a switch would pay off. The matched closed-loop ablation below tests both. It confirms the first—the overrides are inert there too—and contradicts the second for the two fixed paradigms it registered in advance, which are decisively worse in the closed loop. Fixed spatial-greedy, which this table promotes, was not part of the registered design; Section VII-E reports it separately as an exploratory arm.

a) What the geodesic ETA contributes.: Every arm above prices assignments with the geodesic-ETA oracle, so none of them says anything about the oracle itself. We isolate it with a second paired A/B on the same plane. The allocator’s *cost* oracle is switched between geodesic ETA and the Euclidean reference over 500 common seeds, while the *execution* oracle stays geodesic in both arms, so the two are scored against identical ground truth. The three baselines come out bit-identical across the arms, as they must: no baseline reads the flag. The geodesic arm also reproduces the published campaign of Table V to nine decimals, so the comparison is against the configuration the paper actually reports. The oracle helps in every scenario: +2.13 pp under robot failure, +1.83 pp under mixed stress, and +4.10 pp under deadline pressure, each with $p < 10^{-4}$ over 500 pairs. The effect sizes are modest. Cliff’s δ is negligible in the first two scenarios (0.131 and 0.128) and small in the third (0.275). The ordering is the one the mechanism predicts—an accurate arrival estimate matters most where time slack is scarcest—so we report the oracle as a consistent but small gain, not as the main carrier of the result.

E. Matched closed-loop ablation

The proxy result above prices no congestion, replanning latency, or recovery time, so it cannot say whether a switch pays off where those costs are real. We therefore repeated the ablation in the closed loop, holding every non-selector component

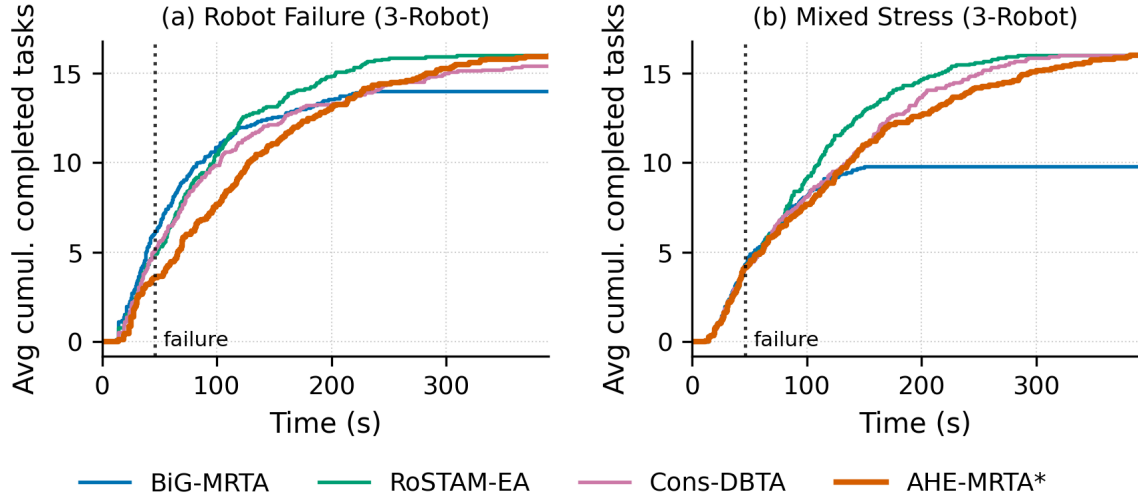


Fig. 12. Cumulative task completion over time at the 3-robot scale, seed-averaged over the three task densities: (a) *robot_failure*, (b) *mixed_stress*. AHE-MRTA completes the full task set in both scenarios. Its makespan is level with the reactive baselines under *robot_failure* (220 s versus 224 s for Consensus-DBTA) but carries a re-assignment penalty under *mixed_stress* (243 s versus 219 s for Consensus-DBTA and 182 s for RoSTAM-EA). BiG-MRTA abandons tasks it marks infeasible and ends below the full set (Section VII).

TABLE XII
MATCHED CLOSED-LOOP ABLATION AT THE PRIMARY 5-ROBOT / 25-TASK SCALE: FOUR ARMS OVER 20 COMMON SEEDS PER SCENARIO, PRE-REGISTERED BEFORE THE CAMPAIGN (PAPER/PREREGISTRATION.MD). PRIMARY ENDPOINT IS EFFECTIVE ON-TIME COMPLETION $CR \times (1 - DVR)$ UNDER *deadline_pressure*; p IS A PAIRED WILCOXON SIGNED-RANK TEST AGAINST *full*, BONFERRONI-CORRECTED WITHIN THE SCENARIO FAMILY, AND δ IS CLIFF’S DELTA WITH A BOOTSTRAP 95% CI. THE OTHER TWO SCENARIOS ARE DESCRIPTIVE.

	full	no-override	fixed-EDF	fixed-orphan
<i>deadline_pressure</i> – primary endpoint				
effective on-time	0.650	0.641	0.556	0.540
p (Bonferroni)	–	1.000	<0.005	<0.005
Cliff’s δ vs full	–	+0.06	+0.57	+0.62
<i>robot_failure</i> – descriptive				
effective on-time	0.986	0.998	0.846	0.850
completion rate	1.000	1.000	0.998	1.000
recovery time (s)	83.7	63.3	88.5	56.6
<i>mixed_stress</i> – descriptive				
effective on-time	0.616	0.606	0.590	0.601

fixed: four arms —*full*, *no-override* ($\arg \max_i D_i$ only), and the two fixed paradigms the selector actually reaches—over three scenarios and 20 common seeds at 5 robots / 25 tasks, 240 runs. Hypotheses, arms, endpoints and the statistical plan were registered before the campaign started and are archived with the artifact; the analysis code was written and committed while one arm was still empty. No seed was lost, so all 20 pair in every arm.

Table XII splits the question in two, and the two halves answer differently.

Being able to switch matters. Pinning the allocator to either fixed paradigm costs 9–11 percentage points of effective on-time completion under deadline pressure (0.650 against 0.556 and 0.540), with large effect sizes ($\delta = 0.57$ and 0.62) that survive Bonferroni correction ($p < 0.005$). The same ordering appears under robot failure, where the two switching arms reach 0.99 and the two fixed arms 0.85. The proxy ranks these two fixed paradigms below the full selector as well, on the same scenario and endpoint (0.348 against 0.296 and 0.259; Table XI), so the two planes agree in direction. They differ in what they support: the closed-loop penalty is larger for fixed-EDF (9.4 pp against 5.2), and only the closed-loop comparison is paired on common seeds and corrected, which is what turns the ordering into a result.

Which mechanism does the switching does not matter. Removing the context-override cascade entirely changes nothing measurable: 0.641 against 0.650, $p = 1.000$ after correction, $\delta = 0.06$ (negligible). Under robot failure the no-override arm is if anything slightly ahead (0.998 versus 0.986, and 63 s versus 84 s to recover, neither surviving correction). The overrides and the dominance dynamic are therefore substitutes rather than a load-bearing layer and its ornament: either route reaches an adequate paradigm, and the cost of having neither is large.

The paradigm the fallback actually returns. The two fixed arms above were registered because they are the paradigms the

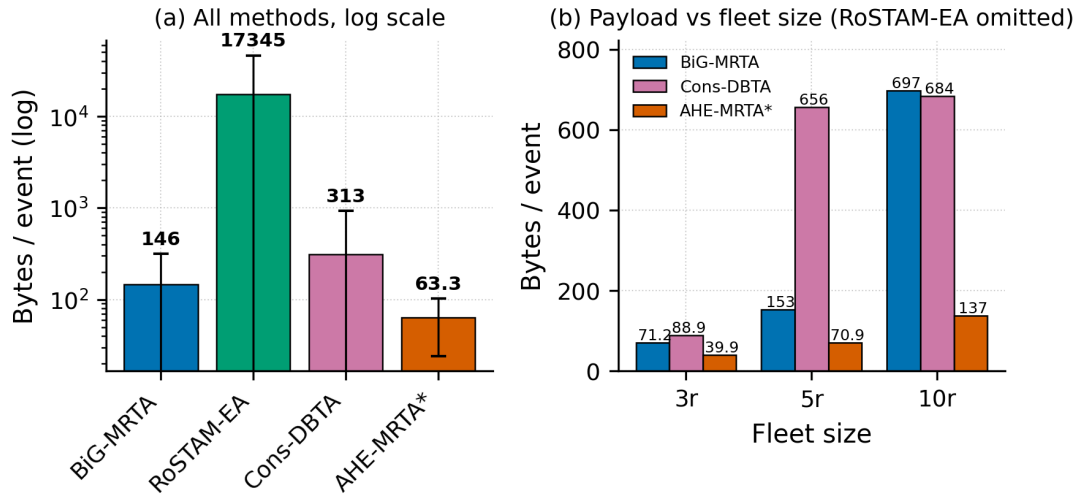


Fig. 13. Communication footprint per allocation event, counted from the fields each protocol transmits. (a) All methods, pooled over scales (log axis). (b) The same payload by fleet size, RoSTAM-EA omitted for range: only per-robot queues are broadcast by AHE-MRTA, so its payload grows linearly in the fleet while bid and utility-matrix exchanges grow with the robot-task product.

overrides target. They are not the paradigm the allocator spends most of its time in: the dominance tier returns spatial-greedy in 99.9% of the states it decides (Section VII-G), and the corrected proxy ablation puts fixed spatial-greedy *ahead* of the full selector (Table XI). Leaving the strongest candidate untested would rest the portfolio claim on two weak alternatives, so we ran it as a fifth arm, matched to the same 20 seeds but outside the registered design and therefore outside the primary Bonferroni family. On the primary endpoint it does not separate from the full selector: 0.638 against 0.650, paired $p = 0.43$, $\delta = +0.09$ (negligible). Under robot failure it is ahead (0.998 against 0.986, raw $p = 0.014$, $\delta = -0.30$) and under mixed stress behind (0.584 against 0.616, raw $p = 0.038$); neither would survive the correction the registered arms passed.

This bounds the portfolio claim more tightly than the registered arms alone would. What the closed loop demonstrates is that pinning the allocator to a paradigm the context does not call for is expensive—fixed-EDF and fixed-orphan lose 9–11 pp—not that the selector beats every fixed policy. The fixed policy that matches the allocator’s own dominant behaviour performs like the full selector, as does dropping the override cascade: 0.650, 0.641 and 0.638 are three descriptions of the same performance. We therefore claim that the portfolio must contain a paradigm suited to the regime and that the allocator must not be pinned away from it; we do not claim that the selector earns its complexity over spatial-greedy in these three scenarios. Separating those two claims needs regimes where spatial-greedy is the wrong answer and the selector has to leave it—which the mixed-stress gap hints at and no experiment here establishes.

Two limits of this experiment are worth stating with the result. The secondary endpoint we registered for *robot failure*, completion rate, turned out to be at ceiling in all four arms (≥ 0.998) and so could not discriminate; effective on-time completion does discriminate there, but we had registered it for the primary scenario only, so it is reported above as descriptive rather than promoted after the fact. And the campaign uses one arena: it closes the matched-ablation gap, not the single-map gap.

F. Communication footprint

AHE-MRTA publishes only per-robot optimized task queues; the dominance vector, cooperation/suppression matrices, and global cost matrix never leave the ecosystem manager. Each method’s payload is counted the same way—the fields its protocol transmits at that event times their field size—so a queue entry costs a 4-byte task id and each robot an 8-byte header, a bid record 16 bytes, and a utility-matrix entry 4 bytes. Figure 13 shows the resulting per-event payload. Pooled over scales it is 63 bytes for AHE-MRTA against 134 and 239 bytes for the commit-once baselines and 1.7×10^4 bytes for RoSTAM-EA, whose population exchange dominates. The pooled figure understates the difference because the interfaces scale differently: AHE-MRTA broadcasts one bounded queue per robot, so its payload grows linearly in the fleet (40, 71, and 137 bytes at 3, 5, and 10 robots), whereas bid and utility-matrix exchanges grow with the robot-task product (BiG-MRTA 71 $\rightarrow$ 697, Consensus-DBTA 89 $\rightarrow$ 684, and RoSTAM-EA $7.4 \times 10^3 \rightarrow 2.6 \times 10^4$ bytes over the same range). We report the payload only and make no claim about its behaviour as a function of switching frequency, which this experiment does not vary.

G. Limitations

We group the limitations by what a reader should do with them: the first group bounds what our evidence establishes, the second bounds where it applies, and the third records artefacts we found in our own instrument and chose to report rather than

repair after the fact.

a) *What the evidence can and cannot support:* (i) **Statistical power varies by scale:** The 5-robot primary scale now carries $n=20$ per cell and supports corrected inference: 20 of 63 cross-method comparisons survive Bonferroni correction, against 0 of 63 when the same cells carried $n=5$. Every method’s point estimates fell when the seed budget was raised—by 0.08–0.16 in effective on-time completion, consistently across all four—so the earlier five-seed estimates were optimistic, though the ranking they implied held. The 3-robot cells carry $n=15$ and the 10-robot cells still carry $n=5$; the latter remain descriptive, and a larger 10-robot budget would extend inference to that scale. The 500-seed proxy planes support their own high-power comparisons. (ii) **Bounded role of the hormone dynamics.** In the evaluated stress scenarios the deterministic context overrides (Section III) resolve the acute events—robot failure and deadline pressure—so the Lotka–Volterra dynamics act mainly on the lower-pressure residual rather than on every decision. Replaying the selector over the 15,058 ecosystem states logged across the 120 AHE-MRTA benchmark runs quantifies the split: the failure override decides 49.2% of states, the deadline override 25.3%, and the dominance tier the remaining 25.5%—where it returns spatial-greedy in 99.9% of cases. Replacing that tier with a constant would change the selected paradigm in 3 of 15,058 states (0.02%); commit-once is never activated in any evaluated scenario and priority-first fires in a single state. We therefore frame the method as a context-driven override cascade *with* an adaptive dynamics tier, not as a dynamics-only selector, and we do not claim the dynamics tier as an operative contributor at these stress levels. The ablation in Table XI (Sec. VII-D) is consistent on the override layer—removing the overrides costs 0.6 pp of navigation-proxy fitness and removing the recovery boost 0.1 pp—but not on the portfolio: the fixed paradigms range from 1.9 pp *above* the selector (spatial-greedy) to 8.2 pp *below* it (commit-once). The closed-loop ablation of item (iii) is what adjudicates between those readings. (iii) **What the closed-loop ablation does and does not settle:** The matched experiment of Section VII-E closes the causal gap this paper previously listed as open—it holds every non-selector component fixed over 20 common seeds—but it settles the portfolio question, not the mechanism question. Switching beats being pinned to a fixed paradigm by a large, corrected margin; the specific selector we propose is *indistinguishable* from the plainer $\arg \max_i D_i$ rule it sits on top of. We therefore claim online switching over a fixed portfolio, and explicitly do not claim that the override cascade is the reason it works. Establishing which switching rules suffice, and whether any is better than the rest, needs a sweep over rules rather than the two points we measured. The experiment also uses a single arena, so it does not close the generalisation gap of item (vi). (iv) **Asymmetric development effort.** AHE-MRTA was developed, and its fixed constants selected, against this arena and these three scenarios, whereas each baseline was implemented once from its source description and not tuned here; a benchmark built around one method favours that method by construction. The selection used holdout seeds, but they are not disjoint from the reported campaign: the sweep that fixed the repair step’s tolerance and reservation gap used seeds 401–420, which lie inside the 1–500 range reported in Section V. We therefore read the four-method comparison as evidence about configured policies in one stack, not about the underlying algorithms.

b) *Scope of the evaluated setting:* (v) **Hardware-bound scale ceiling:** Scaling to 15 robots was infeasible on our 16-thread/16 GiB testbed—multiple per-robot Nav2 stacks failed readiness at launch (startup failures)—so 10 robots is the largest *validated* physical scale. This is a compute limit of the simulation host, not an algorithmic one: in the navigation-independent setting the allocator holds CR at 1.000 and decides in at most 0.215 ms at 10 robots / 50 tasks, and its fitness *rises* with fleet size rather than degrading (Table IV). That bounds the allocator, not the stack it was run in; it says nothing about what a larger Nav2 fleet would do. (vi) **Arena generality:** All experiments use a single layout, so cross-environment generalisation is not shown. The arena is fixed in the placement module; a second layout therefore requires parameterising that module rather than re-tuning the method. We specified and costed that experiment but did not run it here. (vii) **Single-robot capability:** All robots are identical (ST-SR); heterogeneous-capability fleets are not evaluated. The cost function already carries a capability term and the admission gate already rejects incapable pairs, so the extension is a scenario change rather than a method change. We do not claim it until it is measured. (viii) **Ground-truth localization:** The physical benchmark supplies pose through a static `map`→`odom` transform at the spawn pose, which deliberately isolates allocation quality from localization error. Real deployments incur localization drift. The comparison here stays fair, because the contribution is the allocation policy and every method shares the identical ground-truth pose; but quantifying robustness to localization noise (e.g. AMCL) is left to future work.

c) *Model and implementation artefacts:* (ix) **Recovery-time variance:** Recovery time under *mixed_stress* carries high variance (3 σ , AHE-MRTA: 67 ± 78 s over runs spanning 0–263 s). The cause is that some seeds record no in-window failure recovery at all—6 of the 60 runs at 3 robots and 3 of 80 at 5 robots. The *robot_failure* scenario guarantees an in-window failure, so it is the cleaner recovery measure. (x) **Fixed hormone–paradigm mapping:** The A , S , and V matrices are designer-specified; learning them from domain data is left to future work. (xi) **Allocation-fitness model.** The navigation-proxy simulator uses a position-dependent navigation-failure model chosen to *stress* allocation robustness; its failure rate is deliberately higher than the closed-loop Gazebo stack (where Nav2 reaches goals almost always). The proxy therefore measures relative allocation robustness, not an absolute success rate, and is reported alongside—never instead of—the closed-loop Gazebo results. (xii) **Priority is expressed twice in the cost.** The additive term $w_p P_\tau$ discounts high-priority tasks, while the factor $\rho(\pi_\tau)$ of Eq. (8) rescales the whole shaped cost. Replaying the allocator over 2.3×10^4 cost evaluations shows the bracket is negative for every feasible pair—the deadline-capability bonus outweighs the weighted features—so ρ adds about 0.50 to a priority-3 candidate where the additive term grants it only 0.07, reversing the intended ordering. The primary outcomes are priority-

agnostic, and a paired 500-seed A/B with the factor removed moves the priority-weighted proxy fitness by at most 0.8 pp in any scenario (+0.47 pp robot failure, $p=0.046$; +0.77 pp mixed stress, $p=0.0015$; +0.09 pp deadline pressure, $p=0.54$). Every sign is positive: removing the factor helps, and in two of the three scenarios the improvement is statistically detectable at 500 paired seeds. The artefact is therefore small but not invisible in the reported outcomes, and the direction is the one the audit predicted—the factor is a penalty on urgent work, not the discount it was written as. We report the implemented form, and leave the correction to a revision rather than re-tuning the evaluated system after the fact; a corrected ρ would be expected to improve the numbers in this paper by under a percentage point, not to change any ranking. (xiii) **The geodesic ETA is ablated only on the proxy.** All four registered closed-loop arms, and the exploratory fifth, ran with the geodesic oracle enabled, so its contribution is quantified on the navigation proxy alone (Section VII-D). That plane charges nothing for congestion or recovery latency, so the +1.8 to +4.1 pp measured there is not a closed-loop estimate. A matched Gazebo arm with the oracle disabled is the experiment that would settle it; it is specified but not run here.

VIII. CONCLUSION

We presented AHE-MRTA, an event-triggered MRTA framework with fixed, context-driven paradigm selection, a geodesic-ETA cost oracle, and a terminal, bounded load-repair step. All three evaluation layers now put AHE-MRTA first in every scenario, though they disagree on the order below it: commit-once BiG-MRTA is the consistent runner-up on both simulator planes and finishes last in two of the three closed-loop scenarios. On the 500-seed navigation-proxy planes AHE-MRTA leads every scenario, and eight of nine paired comparisons against a baseline are significant; the exception is a tie with commit-once BiG-MRTA under robot failure.

At the primary 5-robot Gazebo scale, 20 seeds per cell support corrected inference: 20 of 63 cross-method comparisons survive Bonferroni correction, 15 in AHE-MRTA’s favour. Within the evaluated map and Nav2 stack, AHE-MRTA reaches full completion under robot failure at 3 and 5 robots and the largest effective on-time completion under deadline pressure—about 12% (5 robots) and 46% (10 robots) above the strongest baseline, whose deadline-violation rate it also cuts by 17% and 49%—at the cost of longer routes, lower workload balance, and decision latency above the commit-once baselines. The 10-robot cells remain descriptive at five seeds.

The matched closed-loop ablation locates the mechanism, and bounds it from both sides. Pinning the allocator to a paradigm the context does not call for costs 9–11 percentage points of effective on-time completion under deadline pressure (corrected $p < 0.005$, large effect), while removing the override cascade changes nothing measurable (0.650 versus 0.641; raw $p=0.59$, $p=1.000$ after correction); an exploratory fifth arm pinned to the paradigm the fallback almost always returns matches the full selector as well (0.638, $p=0.43$). We therefore claim that the portfolio must hold a paradigm suited to the regime and that the allocator must not be pinned away from it—not the particular selector rule we implemented, and not its advantage over its own dominant behaviour. What the present evidence still does not cover is a second arena—every closed-loop number here comes from one map—a 10-robot seed budget large enough to carry correction, and a regime in which spatial-greedy is the wrong answer; those are the natural next experiments.

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